SCALE-COVARIANT ATTENTION FOR TRAINING-FREE HIGH-RESOLUTION DIFFUSION TRANSFORMERS

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ABSTRACT

Generating images above a diffusion transformer's training resolution requires both global composition and local detail, but the appropriate balance differs across attention heads. We introduce Scale-Covariant Attention (SCA), which measures that balance at native resolution and transfers it to larger images. A closed-form output-matching calibration fits one local/global mixing weight per head and noise level, without updating model weights. At inference, subsampled global attention and exact native-extent sliding windows keep attention cost linear in the target token count; a global-only opening establishes layout. On 191 Aesthetic-4K prompts at 4096^2, SCA improves ImageReward over AReA from 0.92 to 1.23 and HPSv2.1 from 27.2 to 29.7, while reducing generation time from 202 to 71 seconds on a GH200. Matched speedups reach $2.99\times$ at 6144^2 and $3.31\times$ at 8192^2. Comparisons also include full attention, DyPE, SEGA, and UltraImage. At 4096^2, SCA improves the shared-protocol baselines; UltraImage's two-stage defaults retain advantages on several preference scores while SCA scores higher on crop MUSIQ. Additional prompts, seeds, and FLUX.2-dev test generality. Paired intervals, ablations, and failure galleries delimit these gains, including mixed results at non-square sizes.

Figure 1: **Global composition and fine detail at 4096^2.** One curated evaluation prompt across six methods: full images above, identical 768×768 crop coordinates below. The portrait and crop were already present in our earlier hero selection. FLUX.1-dev and seed label 0; Full attention, DyPE, SEGA, AReA and SCA share 28 steps and guidance 4.5. *UltraImage uses its two-stage defaults (30+30 steps). This example illustrates detail and artifacts; aggregate comparisons and randomly selected galleries test broader behavior.

1 INTRODUCTION

Text-to-image diffusion transformers such as FLUX.1 (Black Forest Labs, 2024) and Stable Diffusion 3 (Esser et al., 2024) are trained at about one megapixel. At 4096×4096 the number of image tokens grows 16-fold and the cost of full attention about 256-fold, and every attention layer sees a token count and relative positions outside its training distribution. Outputs show duplicated subjects, broken layouts, and blurred texture (He et al., 2023; Du et al., 2023; Issachar et al., 2025;

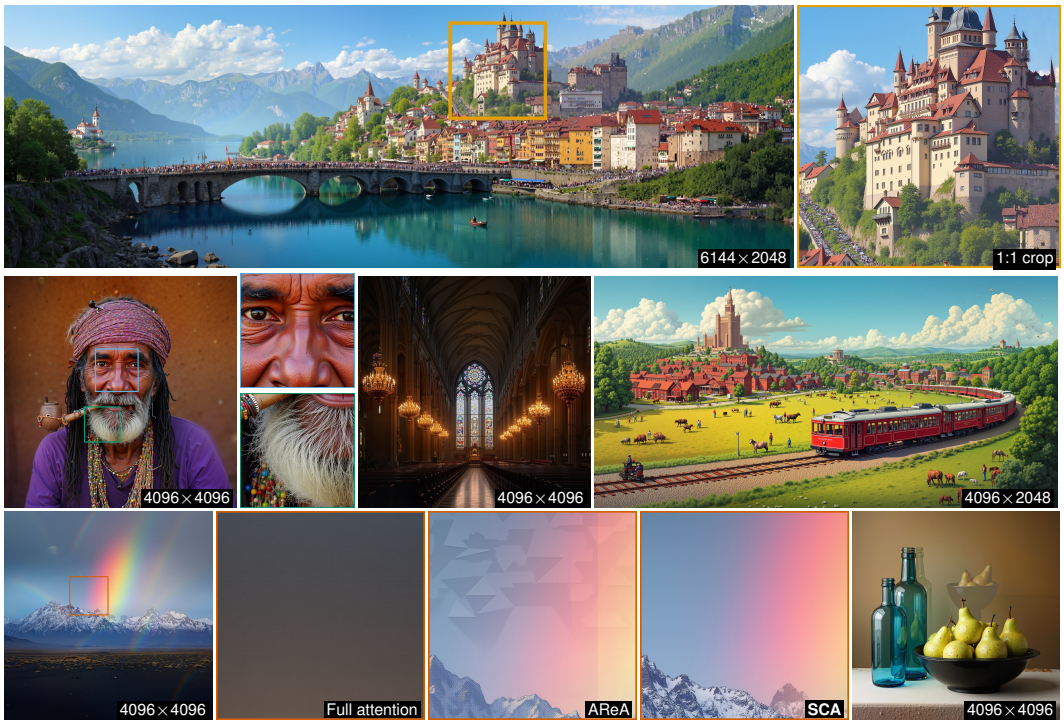

Figure 2: **SCA generates coherent, detailed images up to 6144×2048 from FLUX.1-dev without training.** Unless labeled otherwise, images are SCA outputs at seed 0 (labels give width×height); framed tiles are 1:1 crops at native resolution from the outlined windows. Bottom row: the same prompt and seed at 4096×4096 with full attention, which collapses into a blurred field, with AReA, whose sky breaks into tiled facets, and with SCA.

Rajabi et al., 2026a); under our protocol, full attention in FLUX.1-dev at 4096^2 gives fields of stripes (Figure 2, bottom row).

Training-free fixes edit the positional encoding (Zhuo et al., 2024; Issachar et al., 2025; Zhao et al., 2025; Rajabi et al., 2026b) or restrict the keys each query sees (Zhang et al., 2023; Ma et al., 2025; Xue et al., 2026; Rajabi et al., 2026a). AReA (Rajabi et al., 2026a), the most recent of the second kind, attends to keys subsampled onto a native-size grid for the first 8 of 28 steps, then to windows that shrink to native size, with several repairs for windowing artifacts. AReA shares its receptive-field schedule across heads. Head-specific adaptation is also established: HRDiT assigns per-head scopes under a cost budget (Xue et al., 2026), and UltraImage uses entropy-based head-wise attention concentration (Zhao et al., 2025). We study whether a head's local/global output mixture can be fitted at native resolution and transferred across scales.

At the training resolution, how much of each head's output does a native-size neighbourhood explain, and how much does a downsampled view of the whole image explain? We emulate a target s times larger by shrinking both views by s, so that the ratio of image to window matches the target while full attention stays in distribution and serves as ground truth. A convex mix of the two views fits each head well. Its weight differs strongly between heads, is stable across the tested prompt splits (correlation 0.99), and transfers: the table fitted at emulated scale 2 explains scale 4 with 2.4% more output error than the scale-4 table, and a 1024^2 table explains 1280^2 with 0.8% more (Section 3). The name scale covariance describes the measured transfer, rather than exact invariance: as the image grows, the local view follows a law fixed in tokens, the global view a law fixed in image fractions, and the weight between them stays put.

Scale-Covariant Attention (SCA) turns this measurement into an attention operator (Figure 3). Each head mixes a global branch, which attends to block-centre keys on the native grid with rescaled coordinates, and a local branch, an exact native-extent sliding window, with its calibrated weight. In exact coordinates, both branches keep the native key count and every image-to-image query-key

offset inside the trained range (Proposition 1), their cost is linear in the number of tokens (Proposition 2), and the calibrated weight is the ℓ_2-optimal mix at the emulated scale (Proposition 3). Output matching alone did not select a satisfactory opening phase: applied from the first step at 4096^2, the calibrated mix draws five Moai statues for a prompt asking for one (Figure 8), consistent with a frequency-based explanation: early layout needs global context, while a 64-token window spans a quarter of the image (Argument 1). SCA therefore runs the global branch alone for the first 8 of 28 steps. Our contributions are:

1. **A calibration procedure** for training-free resolution extrapolation: emulate the target scale inside the training distribution and fit each head's receptive-field mix to the model's own full attention in closed form, from statistics collected in one pass. On FLUX.1-dev the mix is head-specific, prompt-stable, and transfers across scale and calibration resolution.
2. **The SCA operator**: a per-head, per-noise-level mix of two branches with native-range image offsets, an opening phase motivated by a frequency argument, and an exact FlexAttention sliding window; one attention layer at 4096^2 takes 18.7 ms against 87.3 ms for full attention.
3. **Five baseline comparisons with paired uncertainty**: comparisons with full attention, AReA, DyPE, and SEGA under a shared FLUX.1-dev protocol, a separate comparison with UltraImage's defaults, and checks on HRGen-1K and FLUX.2-dev. The results distinguish preference scores from fine-detail quality and compute cost.
4. **Ablations** showing the contributions of the opening phase, per-head weights, and block-centre subsampling, with additional seeds, calibration budgets, and gate thresholds to test sensitivity.

2 Scale-Covariant Attention

2.1 Setting and notation

A multimodal diffusion transformer such as FLUX.1 concatenates T text tokens $\mathcal{T}$ with $N = HW$ image tokens on the grid $\mathcal{G} = [H] \times [W]$, $[n] = \{0, \dots, n-1\}$, and applies joint attention in each block (57 blocks of 24 heads in FLUX.1). It is trained at about 1024^2 pixels, a native grid of $n_0 \times n_0$ tokens with $n_0 = 64$. Rotary embeddings give token j a position $p_j \in \mathbb{Z}^2$ (FLUX.1 puts all text tokens at 0) and a rotation R_p with $R_p^\top R_{p'} = R_{p'-p}$. For one head, a pre-rotation query q placed at u, and a set $\mathcal{K}$ of (key, value, key position) triples,

$$\mathcal{A}(q, u; \mathcal{K}) \;=\; \textstyle\sum_{(k,v,u') \in \mathcal{K}} \operatorname{softmax}_{\mathcal{K}}\big(q^\top R_{u'-u}\, k/\sqrt{d}\big)\; v. \tag{1}$$

Full attention is $O_i^{\text{full}} = \mathcal{A}\big(q_i, p_i; \{(k_j, v_j, p_j)\}_{j \in \mathcal{T} \cup \mathcal{G}}\big)$. The target grid is $H = s_H n_0$, $W = s_W n_0$ with integer factors (s when equal), so full attention meets offsets s times the trained range. The latent at noise level σ is $x_\sigma = (1-\sigma)x_0 + \sigma\epsilon$. Softmax attention over a partition of the keys into A and B splits exactly as $O_i^{\text{full}} = m_i O_i^A + (1-m_i) O_i^B$, with m_i the softmax mass on A (Lemma 1): every head is a query-dependent mix of a near and a far field. This identity motivates a mixed receptive field. SCA instead combines two separately normalized operators and fits a per-head weight. Both branches include text keys, and their image key sets can overlap, so the identity is not an exact decomposition of SCA.

2.2 Two operators with native-range image geometry

Global operator. Let $\mathcal{G}_0 = [H/s_H] \times [W/s_W]$ be the $n_0 \times n_0$ grid of blocks. Block-centre subsampling keeps for each block b the token at $\kappa(b) = (s_H b_1 + \lfloor s_H/2 \rfloor,\; s_W b_2 + \lfloor s_W/2 \rfloor)$, and the coarse map $c(p) = (\lfloor p_1/s_H \rfloor, \lfloor p_2/s_W \rfloor)$ sends a position to its block, so $c(\kappa(b)) = b$:

$$\mathcal{K}^{\text{glob}} = \{(k_j, v_j, 0)\}_{j \in \mathcal{T}} \cup \{(k_{\kappa(b)}, v_{\kappa(b)}, b)\}_{b \in \mathcal{G}_0}, \qquad O_i^{\text{glob}} = \mathcal{A}\big(q_i,\, c(p_i);\, \mathcal{K}^{\text{glob}}\big), \tag{2}$$

with $c(p_i)$ replaced by 0 for text queries. This is AReA's first phase with the pooling factor fixed at s. It models content whose extent is a fixed fraction of the image: layout, object count, identity.

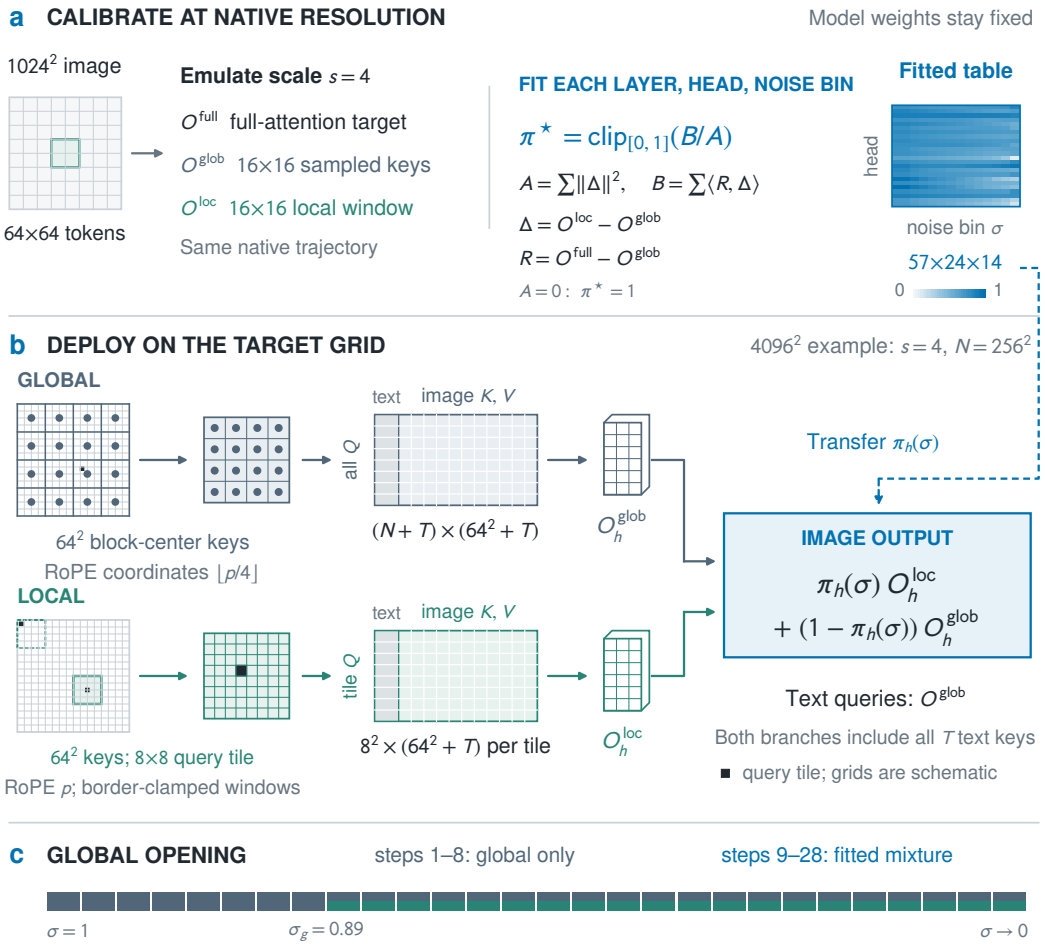

Figure 3: **Calibrate the mixture at native resolution, then transfer it to a larger attention geometry.** (a) On 32 native 1024^2 prompts, SCA emulates scale 4 using 16^2 sampled global keys and a 16×16 local window. Matching the model's full-attention outputs gives one clipped least-squares coefficient per layer, head and noise bin. Sums run over calibration image queries; the model weights remain fixed. The inset shows the archived layer-29 slice in its original head/bin order, not illustrative values. (b) At 4096^2, global attention retains 64^2 block-centre image keys with block-index RoPE. Local attention retains a 64×64 window around each 8×8 query tile with original RoPE. The rectangles show attention support, not measured attention weights; N and T count image and text tokens. Both branches include all text keys. Image queries receive the fitted mixture; text queries use the global output. (c) For the 28-step protocol, the first eight steps use only global attention. The local branch is skipped while $\sigma > 0.89$. Grid drawings are schematic; the coordinate definitions do not assert exact bf16 RoPE equivalence.

Local operator. Image queries form tiles of $t \times t$ tokens ($t = 8$); query i lies in tile $\tau = \lfloor p_i/t \rfloor$. With window side $w = \min(n_0, H)$, the window of tile τ starts at $a(\tau) = \operatorname{clip}\big(t\tau + \lfloor t/2 \rfloor - \lfloor w/2 \rfloor, 0, H - w\big)$ per axis, centred on the tile and clamped to the grid. With $\mathcal{W}(i)$ the $w \times w$ window of i's tile,

$$\mathcal{K}_i^{\mathrm{loc}} = \{(k_j, v_j, 0)\}_{j \in \mathcal{T}} \cup \{(k_j, v_j, p_j)\}_{p_j \in \mathcal{W}(i)}, \qquad O_i^{\mathrm{loc}} = \mathcal{A}\big(q_i,\, p_i;\, \mathcal{K}_i^{\mathrm{loc}}\big), \tag{3}$$

an exact sliding window with the original positions. It models content whose extent is fixed in tokens: edges, texture, fine structure.

Proposition 1 (Trained position range)**.** *Assume exact integer position coordinates and* $t \le n_0 \le \min(H, W)$. *Every query-key offset of the global operator and every image query-key offset of the*

local operator is at most $n_0 - 1$ per axis, and each image query sees $T + n_0^2$ keys in either operator, as in full attention on the native grid.

Proposition 2 (Cost)**.** *Per head and layer, full attention scores $(N+T)^2$ query-key pairs, the global operator $(N+T)(T+N_0)$ with $N_0 = |\mathcal{G}_0| = n_0^2$, and the local operator $N(T+w^2)$. For fixed n_0 and T the total is $O\big(N(T+w^2+N_0)\big)$, linear in N.*

Proofs are in Appendix A. The position bound concerns exact coordinates; the frozen FLUX.1 implementation has bf16 rounding beyond 256 tokens per axis (Appendix C). Native-range geometry does not imply in-distribution hidden states: both operators still act on an extrapolated-resolution trajectory. Text-image offsets in the local operator equal p_i, as in full attention at the target (Remark 1). At 4096^2 SCA scores 7.2 times fewer pairs than full attention, 8.4 times fewer with the opening phase (Table 6).

2.3 Per-head mixing and the opening phase

For an image query of head h in block l at noise level σ,

$$O_i \;=\; O_i^{\mathrm{glob}} \;\text{ if } \sigma > \sigma_g, \qquad O_i \;=\; \pi_{lh}(\sigma)\, O_i^{\mathrm{loc}} + \big(1-\pi_{lh}(\sigma)\big)\, O_i^{\mathrm{glob}} \;\text{ otherwise,} \tag{4}$$

and text queries use O_i^{glob}. The weight $\pi_{lh}(\sigma)$, written $\pi_h(\sigma)$, stands in for the near-field mass m_i; one weight per head is cheap and a few prompts determine it.

Argument 1 (A frequency-based explanation for the opening phase). At noise level σ a spatial frequency band has signal-to-noise ratio $\big((1-\sigma)/\sigma\big)^2 S(f)$ for image spectrum S. If latent signal power decreases with frequency, lower-frequency bands retain more signal at high noise. A window of w tokens spans less than one period below $1/w$, which limits the spatial evidence available in a single attention call. Independently attending to such windows may favor repeated local structures over one large object; a global branch supplies context across windows. Hidden states can already contain information from preceding layers, so this is a mechanistic hypothesis rather than an impossibility result for the full network. A tokenwise squared-error objective can favor local detail without preserving large-scale layout; our generation ablation exhibits this mismatch.

We have not measured the per-band signal-to-noise ratio of FLUX.1 latents, but the argument's prediction holds: a 16-token window gives smaller and more numerous copies than a 64-token one (Figure 8). We set $\sigma_g = 0.89$, the first 8 of 28 steps at maximum shift, which is also AReA's switch point; gating 13 steps gives a 4×4 block mosaic, because all queries of a block share one coarse position.

2.4 Calibration by output matching

The weights must come from where the model is trustworthy, which the target is not. On an in-distribution $n \times n$ grid ($n = 64$ at 1024^2), emulated scale s_c uses subsampling factor s_c, window side n/s_c, and tile side $\mathrm{round}(tn/(s_c n_0))$, so that the ratio of grid to window is s_c, as at a target of scale s_c. The sampler runs unmodified full attention; at every block and step both emulated operators are evaluated on the same queries, keys, and values. For block l, head h, and noise bin β, sums over prompts, steps with $\sigma \in \beta$, image queries, and channels give

$$A = \textstyle\sum \|O^{\mathrm{loc}} - O^{\mathrm{glob}}\|^2, \quad B = \sum \langle O^{\mathrm{full}} - O^{\mathrm{glob}},\, O^{\mathrm{loc}} - O^{\mathrm{glob}} \rangle, \quad E = \sum \|O^{\mathrm{full}} - O^{\mathrm{glob}}\|^2, \tag{5}$$

and the squared error of the mix with weight π is $J(\pi) = E - 2\pi B + \pi^2 A$.

Proposition 3 (Closed-form calibration and transfer cost)**.** *In each cell (l, h, β), $\pi^\star = \mathrm{clip}(B/A, 0, 1)$ ($\pi^\star = 1$ if $A = 0$) minimizes J on $[0,1]$. Any other weight $\pi' \in [0,1]$ (fitted at another scale or resolution, per layer, shared, or a hard switch) has $J(\pi') - J(\pi^\star) \geq A(\pi' - \pi^\star)^2$, with equality when $0 < \pi^\star < 1$.*

So the per-head table has the lowest output error of any rule constant within a cell, and replacing it costs an A-weighted squared distance. This holds at the emulation scale; Section 3 measures empirically how well tables transfer between scales and resolutions. The same sums score every alternative rule without another forward pass. We use 14 bins of width $1/14$ in σ and interpolate linearly between the mean sampled noise levels of occupied bins at inference. One pass collects

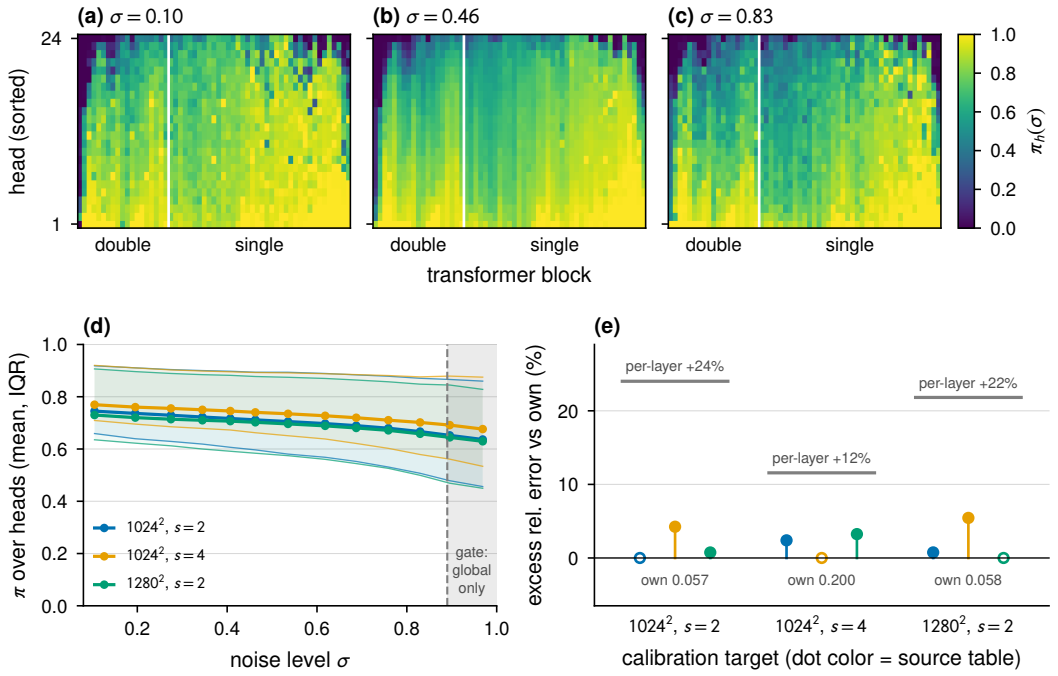

Figure 4: **Mixing weights vary strongly across heads, yet a calibrated table transfers across scale and resolution at little cost.** (a to c) Local weight $\pi_h(\sigma)$ of the scale-4 table used at inference, for all 57 blocks and 24 heads at three noise levels below the gate ($\sigma = 0.89$). Heads are sorted within each block by their mean weight, in the same order in every panel; the white line separates the 19 double-stream blocks from the 38 single-stream blocks. Below the gate, 48.7% of the cells lie in $[0.2, 0.8]$ and the standard deviation over heads is 0.22. (d) Mean of π over all heads, with thin lines and light fill marking the interquartile range, for three calibrations: 1024^2 with 32 prompts at emulated scales 2 and 4, and 1280^2 with 16 prompts at scale 2. In the shaded region the gate replaces the table with the global branch. (e) Relative output error of each per-head table on each calibration target, as excess over the target's own table (hollow dot; its absolute error is printed below). A transferred table adds at most 5.45% error, while the target's own per-layer fit (grey bar) adds 12% to 24%.

emulated scales 2 and 4. The default table uses 32 prompts at 1024^2; 16 prompts at 1280^2 assess transfer. A target with factors (s_H, s_W) uses the table whose scale is nearest $\max(s_H, s_W)$ in log space, here always 4. The local operator is one FlexAttention call (Dong et al., 2024) over tile-major tokens with the window membership as block mask, so the window is exact. Algorithm 1 and implementation details are in Appendix C.

3 What the heads want

We calibrate FLUX.1-dev on 32 prompts at 1024^2 and 16 at 1280^2, written for this purpose and disjoint from the evaluation sets. Errors are relative, $\sum J / \sum \|O^{\text{full}}\|^2$, pooled over blocks, heads, and steps.

Heads differ. Over the 1368 heads, the standard deviation of $\pi_h(\sigma)$ is 0.24 to 0.26 at every noise level (emulated scale 2); half of all (head, noise level) cells lie in $[0.2, 0.8]$ (50.4%; 49.7% at scale 4), 11.7% exceed 0.95, and 2.9% are below 0.05 (Figure 4). The mean weight is 0.64 to 0.74 (0.68 to 0.77 at scale 4): the typical head leans local but still draws on the global branch.

The table is stable across tested prompt splits. Tables fitted on disjoint halves of the prompts correlate at 0.989 over all cells (0.986 at scale 4; 0.974 for 16 prompts at 1280^2), and each half's table scored on the other half gives a relative error of 0.0576 against 0.0574 in sample. Tables

Table 1: **Evaluation protocol.** The FLUX.1-dev attention comparisons share sampler settings and prompt seeds. UltraImage uses a separate two-stage protocol; FLUX.2-dev uses its own native-resolution schedule (Appendix G). Prompt counts exclude four development prompts. Appendices D and E give hyperparameters and metric definitions.

Models	FLUX.1-dev (Black Forest Labs, 2024) (main results); FLUX.1-Krea-dev (Black Forest Labs & Krea AI, 2025) [TODO: E9]; FLUX.2-dev (Black Forest Labs, 2025) (50 prompts); bf16.
Sampler	28 Euler flow-matching steps, guidance 4.5, shift $\mu = 1.15$ at every size, seed label 0 per prompt, tiled VAE decoding; UltraImage and FLUX.2 exceptions are detailed below.
Sizes	4096^2, 2048×4096, 2048×6144 (191 prompts each); 6144^2 (100 prompts, [TODO: E2]); 8192^2 (12 exploratory paired prompts, metrics and qualitative examples).
Prompts	Aesthetic-4K evaluation set with 4K references (Zhang et al., 2025a), 191 prompts, fixed 46-prompt ablation subset; HRGen-1K (Rajabi et al., 2026a), 200 prompts.
Baselines	Full attention; AReA, released code (Rajabi et al., 2026a); DyPE (Issachar et al., 2025), SEGA (Rajabi et al., 2026b), UltraImage (Zhao et al., 2025) (two-stage defaults).
Metrics	ImageReward, HPSv2/2.1, PickScore, CLIP, aesthetic; FID, KID, IS, patch FID/KID; MUSIQ (full, 1024^2 crops), CLIP-IQA, ARNIQA; 95% paired bootstrap intervals over prompts.
Hardware	One NVIDIA GH200 per process; verified timing logs identify the GH200 120GB model. Clean latency uses batch 1; campaign timings are marked separately.

fitted to 4, 8, or 16 prompts also give similar generation metrics on the 46-prompt evaluation subset (Section 4.3); this tests generation sensitivity, rather than proving that the tables are identical.

The weights transfer across the tested scales and resolutions. The scale-2 table applied at scale 4 gives a relative error of 0.205 against 0.200 for the scale-4 table, still below every shared rule fitted at scale 4. The 1024^2 table scored at 1280^2 gives 0.0589 against 0.0585. Over all pairs of calibration targets, a transferred table adds at most 5.45% error, while each target's own per-layer fit adds 12 to 24%. The two resolutions emulate scale 2 with windows of 32 and 40 tokens. Transfer between them is consistent with weights depending primarily on the image-to-window ratio over these tested settings.

Per-head beats coarser rules; output error misses layout. Relative to the best hard switch, per-head mixing lowers the relative error from 0.092 to 0.057 at scale 2 and from 0.261 to 0.200 at scale 4; per-layer tables reach 0.071 and 0.223, one shared weight per noise level 0.077 and 0.230 (Figure 10). In output energy the best hard switch never switches (Appendix B); the opening phase overrides this on purpose (Argument 1).

4 Experiments

Protocol. The FLUX.1-dev attention comparisons use AReA's sampler settings, and AReA runs from its released code with FLUX.1-dev (Appendix M). SCA uses $t = 8$, $w = 64$, $\sigma_g = 0.89$, and the scale-4 table from 32 prompts at 1024^2. "SCA + band guard" was our first full configuration (Appendix L). Our UltraImage comparison uses its released guided two-stage recipe (30 steps at each stage, guidance 3.5 then 7); it is a system-level comparison with a different sampling budget. FLUX.2-dev is evaluated separately.

4.1 Main results

At 4096^2, SCA beats AReA on every preference, alignment, and no-reference quality metric (Table 2): ImageReward $+0.31$ $[0.23, 0.39]$, HPSv2.1 $+2.50$ $[2.14, 2.91]$, PickScore $+0.73$ $[0.62, 0.85]$, CLIP score $+1.45$ $[1.10, 1.81]$, MUSIQ_c $+10.9$ $[9.5, 12.3]$. These scalar intervals use the same 2000 paired prompt draws as Figure 5. FID falls by 7.6 $[2.9, 12.8]$ under the 1000-draw reference-image-cluster bootstrap. Corrected patch FID, patch KID, whole-image KID, and Inception score have intervals spanning zero; in particular, patch KID is not significant after correcting reference crops and grouping repeated image identities (Appendix E). SCA takes 71.1 s per image, giving a $2.6\times$

Table 2: **Six methods at 4096^2 on 191 paired prompts.** Higher is better except FID; HPSv2.1 is $\times 100$. The first block shares FLUX.1-dev, 28 steps, guidance 4.5 and prompt seeds; bold marks its best mean. UltraImage is a separate system-level comparison with its two-stage defaults. Paired uncertainty appears in Figure 5 and Appendix H; all metrics, aspect ratios and timing provenance remain in Table 8.

Method	ImageReward	HPSv2.1	CLIP	MUSIQ$_c$	FID↓
Full attention	−2.18	7.61	15.38	33.2	312.7
DyPE	−0.11	20.52	24.84	50.4	206.2
SEGA	0.55	24.37	26.93	50.4	165.4
AReA	0.92	27.20	27.67	50.4	146.0
SCA	**1.23**	**29.71**	**29.11**	**61.2**	**138.4**
Two-stage defaults: 30+30 steps, guidance 3.5 then 7					
UltraImage	1.28	30.80	28.80	41.8	142.7

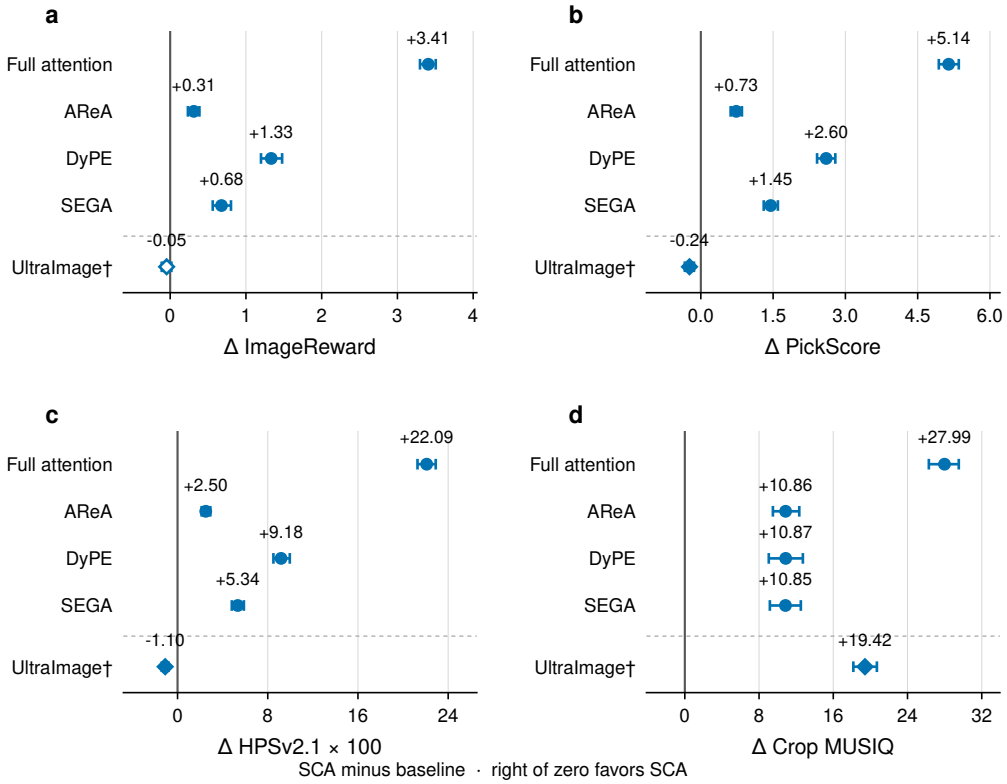

Figure 5: **Paired quality differences at 4096^2.** Each point is SCA minus the named baseline on the same 191 held-out prompts; positive values favor SCA. Whiskers are 95% percentile intervals from 2000 paired prompt bootstrap draws, with no multiplicity correction. All methods use generation seed label 0, so these intervals describe prompt sampling uncertainty, not variability across generation seeds. The open marker indicates an interval containing zero. Full attention, AReA, DyPE, and SEGA use the common FLUX.1-dev sampler (28 steps, guidance 4.5). DyPE and SEGA are not evaluated on their preferred Krea checkpoint here. †UltraImage uses its separate two-stage recipe, with 30 steps per stage and different guidance; it is not an equal-cost comparison. UltraImage scores higher on PickScore and HPSv2.1, while SCA scores higher on crop MUSIQ; their ImageReward difference is unresolved.

speedup over full attention and $2.85\times$ over AReA. Full attention collapses into stripes and is significantly worse than AReA on every reported per-image preference, alignment, and quality metric. The paired crops expose smoothing, speckle, and tiling artifacts (Figures 1 and 6).

Table 3: **Seconds per image**, batch 1, idle GH200, 28 steps with text encoding and VAE decoding, median of three after warm-up. All populated cells use the continuing E3 sweep. Full attention at 8192^2 is outside this sweep. **?**: pending.

Size (N tokens)	Full attention	AReA	SCA
2048^2 (16,384)	22.9	41.0	**18.2**
3072^2 (36,864)	73.9	114.3	**40.2**
4096^2 (65,536)	186.5	202.4	**71.1**
6144^2 (147,456)	**?**	482.3	**161.6**
8192^2 (262,144)	not run	952.0	**287.6**

Table 4: **Generality and sensitivity** at 4096^2: paired differences from the stated reference; HPSv2.1 is $\times 100$. Calibration and gate variants use default SCA (32 calibration prompts, $\sigma_g = 0.89$) as reference. n counts evaluation prompts. Intervals are in Appendix H. **?**: pending.

Setting	n	Reference	ΔIR	ΔHPSv2.1	ΔMUSIQ$_c$
HRGen-1K	200	AReA	$+0.642$	$+4.48$	$+12.71$
Seed 1	46	AReA	$+0.305$	$+2.19$	$+11.58$
Seed 2	46	AReA	$+0.355$	$+2.80$	$+11.04$
4 calibration prompts	46	SCA	$+0.012$	-0.02	$+0.09$
8 calibration prompts	46	SCA	$+0.010$	$+0.01$	-0.03
16 calibration prompts	46	SCA	-0.007	-0.01	-0.11
Gate $\sigma_g = 0.85$	46	SCA	-0.015	-0.85	-2.04
Gate $\sigma_g = 0.93$	46	SCA	-0.004	-0.15	$+0.65$
FLUX.2-dev	50	Full attention	$+0.456$	$+1.84$	$+21.44$
FLUX.1-Krea-dev	**?**	AReA	**?**	**?**	**?**

Broader comparisons. SCA also improves every reported per-image preference, alignment, and quality metric over DyPE and SEGA under the shared protocol (Figure 5 and Appendix H). These comparisons hold the FLUX.1-dev backbone and sampling settings fixed; they do not reproduce each baseline's separately tuned defaults. UltraImage presents a tradeoff: its HPSv2.1 is higher by 1.10 $[0.80, 1.40]$ and PickScore by 0.24 $[0.14, 0.33]$, while its ImageReward advantage of 0.047 $[-0.016, 0.111]$ is not resolved by the interval. SCA has higher CLIP score by 0.31 $[0.01, 0.63]$ and crop MUSIQ by 19.4 $[18.2, 20.7]$, and lower FID and patch FID. Campaign means are 71.5 s for SCA and 308.5 s for UltraImage, including its per-prompt preparation; these are not isolated-node benchmarks. Additional sampling-budget checks are in Appendix I.

Other aspect ratios. The two non-square settings differ in which metrics favor SCA. At 2048×4096 (factors 2 and 4) SCA has mixed results: ImageReward (-0.00 $[-0.08, 0.07]$) and FID have intervals spanning zero, while the two MUSIQ scores and HPSv2 favor SCA and aesthetic score and CLIP-IQA favor AReA. Corrected patch FID and KID also favor AReA: patch FID is worse by 5.5 $[1.9, 9.5]$ and patch KID ($\times 10^3$) by 1.95 $[0.39, 3.70]$. At 2048×6144 (factors 2 and 6) SCA is significantly better on ImageReward ($+0.27$ $[0.15, 0.40]$), FID (-19.8 $[-24.4, -11.8]$), KID, Inception score, PickScore, HPSv2, both MUSIQ variants, CLIP-IQA, and ARNIQA, and significantly worse on aesthetic score (-0.15). Corrected patch FID and KID point estimates favor AReA at this size, but neither difference is resolved: patch FID $+3.6$ $[-0.2, 11.3]$ and patch KID ($\times 10^3$) $+1.70$ $[-2.22, 6.07]$. The scale-2 table changes the 2048×4096 results negligibly (Appendix K). Figure 19 shows four examples sampled from the completed 12-prompt exploratory 8192^2 run; grid and stripe artifacts remain. The paired 8K scores favor SCA on preference and alignment, while the MUSIQ differences remain unresolved (Appendix J). [TODO: E2: 6144^2 metrics.]

Latency. SCA's attention is linear in N (Proposition 2). At 4096^2 one attention layer takes 18.7 ms against 87.3 ms for full attention, and a whole image 71.1 s against 186.5 s (Table 3 and Figure 7). The continuing sweep gives 18.2 s at 2048^2 and 40.2 s at 3072^2 for SCA, compared with 41.0 and 114.3 s for AReA. At 6144^2 and 8192^2, SCA takes 161.6 and 287.6 s. From 2048^2 to 8192^2, pixel

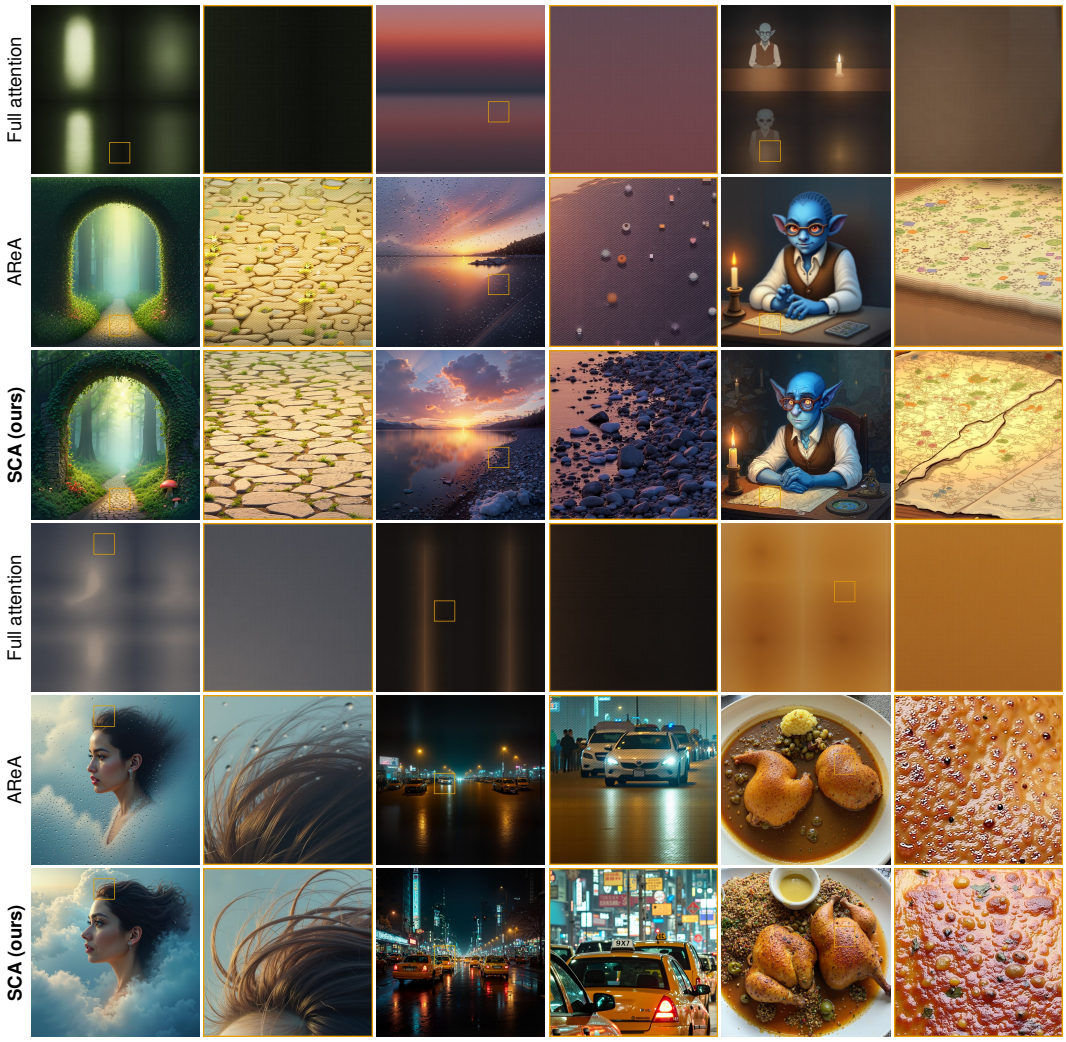

Figure 6: **At 4096×4096, full attention collapses, AReA loses detail and adds texture artifacts, and SCA keeps both layout and fine detail.** FLUX.1-dev, same prompt and seed (0) per column pair; each full image is followed by a 1:1 crop of its outlined 512×512 window. Crop windows are chosen automatically as the region with the most edge energy averaged over the three methods. Prompts were picked for variety from the 191 evaluation prompts; uncurated random samples are in Figure 12.

count grows 16× and measured time grows 15.8×. [TODO: E3: finish the full-attention 6144^2 timing group.]

4.2 Ablations

Table 5 changes one component at a time (examples in Figure 16). **The opening phase is essential:** without it (v0), ImageReward falls by 1.33 $[1.06, 1.60]$ and subjects repeat; CLIP-IQA and ARNIQA rise, since each copy is sharp. **The per-head mix carries global context after the opening phase:** local-only attention in the last 20 steps lowers ImageReward by 0.36 $[0.21, 0.52]$, HPSv2.1 by 2.86, PickScore by 1.20, and CLIP score by 1.93. **Per-head weights beat shared and per-layer weights, in the order output error predicts:** one shared weight per noise level (mix error 0.230) lowers ImageReward by 0.09 $[0.01, 0.17]$ and MUSIQ_c by 10.4; per-layer weights (0.223) still lose significantly on HPSv2.1, PickScore, ARNIQA, and MUSIQ_c. **Neither branch works alone, and block centres beat block means:** global only loses on every metric, and mean pooling blurs the

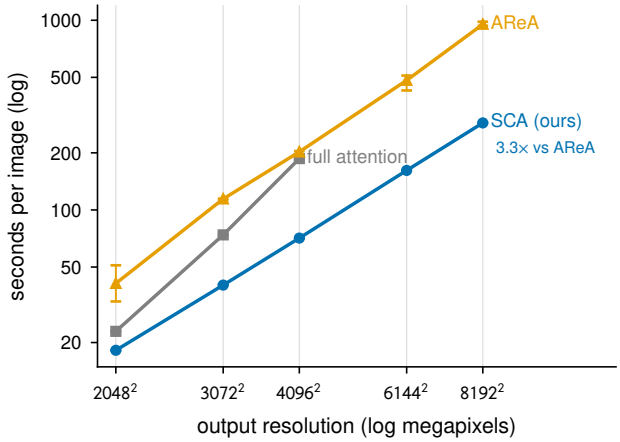

Figure 7: **Measured generation latency across resolutions.** Seconds per image at batch size 1 on an idle GH200, including text encoding and VAE decoding. Points are medians of three prompts after a generated warm-up in the same process; whiskers show the observed minimum and maximum, not confidence intervals. The verified sweep reaches 8192^2 for SCA and AReA and 4096^2 for full attention. At 4096^2, SCA takes 71.1 s, full attention 186.5 s, and AReA 202.4 s. At 6144^2 and 8192^2, SCA takes 161.6 and 287.6 s versus AReA's 482.3 and 952.0 s, giving $2.99\times$ and $3.31\times$ speedups.

Table 5: **Ablations at** 4096^2, 46 prompts. All metrics and intervals remain in Table 11. First row: absolute values for SCA final; other rows change one thing and give the paired difference. **Bold blue** / *orange italic*: significantly better / worse. HPSv2.1 and CLIP-IQA are $\times 100$. "Mix err.": relative output error of the weight rule at emulated scale 4.

Variant	Mix err.	IR	HPSv2.1	PickS.	CLIP-IQA	MUSIQ_c
SCA final (reference)	0.200	1.12	29.98	22.96	71.3	62.5
w/o per-head mixing ($\pi = 1$)	0.261	-0.36	-2.86	-1.20	-4.6	-0.3
one shared $\pi(\sigma)$	0.230	-0.09	-1.83	-0.38	-7.7	-10.4
per-layer $\pi_\ell(\sigma)$	0.223	-0.05	-0.70	-0.16	-0.9	-2.4
global only ($\pi = 0$)	0.627	-0.40	-5.66	-1.44	-27.0	-11.7
mean pooling	n/a	-1.52	-9.75	-2.60	-10.1	-4.2
v0: no opening phase	0.200	-1.33	-4.07	-2.18	$+2.9$	$+2.5$

global keys (ImageReward -1.52). Band-guard ablations, including hard per-head routing, are in Appendix L.

4.3 Generality and sensitivity

On the first 200 HRGen-1K prompts, SCA improves ImageReward over AReA from 0.138 to 0.780, with a paired gain of 0.642 $[0.539, 0.755]$, HPSv2.1 by 4.48 $[3.96, 4.98]$, and crop MUSIQ by 12.7 $[11.2, 14.3]$. All ten per-image metrics improve; there are no reference images for distributional evaluation. On the 46-prompt Aesthetic-4K subset, seeds 1 and 2 retain ImageReward gains of 0.305 $[0.172, 0.458]$ and 0.355 $[0.244, 0.470]$ over AReA. These are two additional seeded runs on the subset, not a multi-seed rerun of all 191 prompts.

Reducing calibration to 4, 8, or 16 prompts changes ImageReward by at most 0.012 on this subset, with intervals spanning zero. No tested per-image metric differs significantly for the four-prompt table. This suggests that modest calibration data can suffice here; it does not establish equivalence for other prompt distributions. Gate threshold 0.85 lowers HPSv2.1 by 0.85 and crop MUSIQ by 2.04, whereas 0.93 leaves these metrics unresolved but lowers CLIP and PickScore. The three thresholds therefore do not identify a metric-independent optimum.

With a new table calibrated on 16 native-resolution prompts, SCA improves FLUX.2-dev over full attention at 4096^2 on 50 evaluation prompts: ImageReward rises from 0.748 to 1.204, HPSv2.1 from 26.90 to 28.74, and crop MUSIQ from 38.5 to 60.0. CLIP's interval spans zero; the other nine per-image metric intervals favor SCA. Mean denoising-plus-decoding time falls from 444.0 to 220.2 s, with text embeddings precomputed for both methods. This extends the full-attention comparison to a second model; it does not establish an advantage over other extrapolation methods on FLUX.2. Protocol details and all paired intervals are in Appendices G and H. [TODO: E9: FLUX.1-Krea-dev.]

5 Related work

Training-free high resolution. Earlier methods change convolutions, spectra, or attention temperature (He et al., 2023; Zhang et al., 2023; Huang et al., 2024; Jin et al., 2023), fuse global and patch paths (Du et al., 2023; Yang et al., 2024; Qiu et al., 2024), or guide with a native pass (Du et al., 2024; Bu et al., 2025; Koh et al., 2025). Positional methods adapt RoPE for extrapolation (Zhuo et al., 2024; Issachar et al., 2025; Zhao et al., 2026; Rajabi et al., 2026b; Liu et al., 2026). UltraImage combines dominant-frequency correction with entropy-derived attention concentration per head (Zhao et al., 2025); its observation of stable local/global head roles precedes ours. ResDiT combines position-scaled global attention with native-size patch attention through patch-wise spectral fusion and uses global-only, fused, and local-only sampling phases (Ma et al., 2025). HRDiT chooses per-head scopes with an image-loss sensitivity estimate and a cost-constrained integer program (Xue et al., 2026). AReA schedules pooled global attention followed by shrinking windows (Rajabi et al., 2026a), while FreeScale fuses attention outputs by frequency band (Qiu et al., 2024). Against this prior work, SCA studies a specific calibration procedure: emulating target scale at native resolution, fitting a per-head/per-noise output mixture in closed form, and transferring that mixture to an operator with bounded key counts. Head specialization, two branches, and a global opening phase are established ingredients.

Efficient attention. ToDo attends to downsampled keys without rescaling coordinates (Smith et al., 2024); DiTFastAttn, Sparse VideoGen, Sliding Tile Attention, and Sparse-vDiT pick windows or sparsity per layer or head at trained resolutions (Yuan et al., 2024; Xi et al., 2025; Zhang et al., 2025b; Chen et al., 2025); our local branch is neighbourhood attention (Hassani et al., 2022). SelfExtend groups distant positions (Jin et al., 2024), MoA assigns per-head spans (Fu et al., 2024), and DuoAttention separates retrieval heads from local ones (Xiao et al., 2024). HiLo (Pan et al., 2022) and Shunted attention (Ren et al., 2021) train vision transformers whose heads split fine and coarse keys by design. Our calibration adds measurements of output-mixture transfer to this existing line of head-specific attention methods.

6 Limitations and conclusion

At 2048×4096, SCA has no resolved ImageReward advantage over AReA and loses on several other metrics. Small extra figures can still appear late in sampling, the exploratory 8K gallery retains grid artifacts, and one development prompt keeps doubled limbs in every variant (Appendix O). The opening phase was selected from development generations; the output-error objective alone did not select it. Transfer across emulated scales supports, but cannot prove, that the weights depend only on the ratio of image to window. The full 191-prompt comparison uses FLUX.1-dev and one seed per prompt; extra seeds cover 46 prompts, and the second-model comparison covers 50 prompts against full attention. UltraImage has higher preference scores under its more expensive two-stage recipe, so SCA does not dominate all systems or metrics. We have not yet completed direct HRDiT, SigMa or ResDiT baselines, and our guided UltraImage configuration does not cover its direct-extrapolation variant; superiority over those untested settings is unestablished. The low-sample FID estimates and automatic preference models cannot replace human evaluation, and the intervals are unadjusted for multiple comparisons. [TODO: E2, E3, E9: the 6144^2 comparison, remaining large-resolution latency measurements, and Krea results remain pending.]

Each head's split between a neighbourhood fixed in tokens and a view of the whole image can be measured once at the training resolution and reused at 4K; with it and an opening phase, SCA beats AReA's released code at 4096^2 in a third of its time.

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A PROOFS

Lemma 1 (Softmax split)**.** *Let the keys of a query be partitioned into A and B, with logits ℓ_j. Then $O^{\text{full}} = m\,O^A + (1-m)\,O^B$ with $m = \sum_{j\in A} e^{\ell_j} / \sum_j e^{\ell_j}$ and O^A, O^B the softmax outputs restricted to A and B.*

Proof. Write $Z_A = \sum_{j\in A} e^{\ell_j}$ and Z_B likewise. Then $O^{\text{full}} = (\sum_A e^{\ell_j} v_j + \sum_B e^{\ell_j} v_j)/(Z_A + Z_B) = \frac{Z_A}{Z_A+Z_B} O^A + \frac{Z_B}{Z_A+Z_B} O^B$. □

Proof of Proposition 1. Global operator. Image queries sit at $c(p_i) \in \mathcal{G}_0 = [n_0]^2$ and text queries at $0 \in \mathcal{G}_0$; subsampled keys sit at $b \in \mathcal{G}_0$ and text keys at 0. Any two points of $[n_0]^2$ differ by at most $n_0 - 1$ per axis. The key count is $T + |\mathcal{G}_0| = T + n_0^2$.

Local operator. Fix one axis of length $H \geq w$ and a tile $[t\tau,\, t\tau + t)$ with $t \leq w$. The unclamped start $a' = t\tau + \lfloor t/2 \rfloor - \lfloor w/2 \rfloor$ satisfies $a' \leq t\tau$ because $\lfloor t/2 \rfloor \leq \lfloor w/2 \rfloor$, and $a' + w \geq t\tau + t$ because $w - \lfloor w/2 \rfloor = \lceil w/2 \rceil \geq \lceil t/2 \rceil = t - \lfloor t/2 \rfloor$. If $a' < 0$ the window is $[0, w)$, and $w \geq a' + w \geq t\tau + t$. If $a' > H - w$ the window is $[H - w, H)$, with $H - w < a' \leq t\tau$ and $H \geq t\tau + t$. In every case the window contains the tile, so it contains each query of the tile, and two positions in a window of side $w \leq n_0$ differ by at most $n_0 - 1$. The key count is $T + w^2 = T + n_0^2$ when $n_0 \leq \min(H, W)$. □

Remark 1 (Text-image offsets). FLUX.1 places all text tokens at position 0, so the offset between an image query at p_i and a text key is p_i. In the global operator the query sits at $c(p_i) \in [n_0]^2$, inside the trained range. In the local operator it keeps p_i, up to $sn_0 - 1$ per axis, exactly as in full attention and AReA's windows at the target. Mapping text offsets through $c(\cdot)$ in the local operator is a possible refinement that we have not tested.

Proof of Proposition 2. Full attention scores every query against every key, $(N + T)^2$ pairs. The global operator scores $N + T$ queries against $T + |\mathcal{G}_0|$ keys with $|\mathcal{G}_0| = N/(s_H s_W) = n_0^2$. The local operator scores N image queries against T text keys and w^2 window keys. The sum is at most $(2N + T)(T + w^2 + N_0)$, which is $O(N(T + w^2 + N_0))$ for fixed n_0, T. Value aggregation has the same pair counts. In the opening phase the local term is absent. The FlexAttention kernel visits whole blocks and masks the extra pairs: a query block of 8×16 tokens spans two tiles whose windows are 8 columns apart, so it visits a 64×72 key region, 12.5% more than one window, plus partial blocks at clamped borders. This changes the constant, not the order. □

Proof of Proposition 3. $J(\pi) = E - 2\pi B + \pi^2 A$ with $A \geq 0$. If $A = 0$ then $O^{\mathrm{loc}} = O^{\mathrm{glob}}$ on every summand, so $B = 0$ and J is constant; any π, in particular $\pi^\star = 1$, is optimal and the bound holds trivially. If $A > 0$, J is a strictly convex quadratic with unconstrained minimizer B/A, and its minimizer on $[0, 1]$ is the projection $\mathrm{clip}(B/A, 0, 1)$. For any π',

$$J(\pi') - J(\pi^\star) = A(\pi' - \pi^\star)^2 + 2(\pi' - \pi^\star)(A\pi^\star - B).$$

If $0 < \pi^\star < 1$ then $A\pi^\star = B$ and the second term vanishes. If $\pi^\star = 0$ then $B \leq 0$, so $A\pi^\star - B \geq 0$ and $\pi' - \pi^\star \geq 0$. If $\pi^\star = 1$ then $B \geq A$, so $A\pi^\star - B \leq 0$ and $\pi' - \pi^\star \leq 0$. In both clipped cases the second term is nonnegative.

Any rule that assigns one weight per cell, such as a per-layer table, a weight shared per noise level, a hard switch, or a table fitted elsewhere, is a feasible π' in every cell, so its total error is at least the per-head table's, by $\sum_{\text{cells}} A(\pi' - \pi^\star)^2$ or more. A table interpolated in σ between the mean sampled noise levels of occupied bins is not constant within a bin; the statement applies to the bin-constant table used to score rules in Section 3. □

The frequency argument in more detail. Argument 1 rests on two facts and one assumption. The first fact is that the noisy latent is a sum of signal and independent white noise, so a band's signal-to-noise ratio scales as $((1 - \sigma)/\sigma)^2 S(f)$; at $\sigma = 0.89$ the factor is 0.015, and at $\sigma = 0.5$ it is 1. The second is that a window of w tokens cannot distinguish a component of period longer than w from a smooth trend within the window, so windows cannot coordinate a layout at that scale except through what earlier layers mixed into their keys. The assumption is that natural-image latent spectra fall with frequency, which we have not measured for FLUX.1 latents. Together they predict that windowed attention fails first and worst at high noise and for objects larger than the window, which matches the duplication we observe (Figure 8) and the effect of a 16-token window.

B Calibration details and statistics

Statistics. For every emulated scale, block, head, and noise bin, the pass stores A, B, E, the error of the local branch alone $\sum \|O^{\mathrm{full}} - O^{\mathrm{loc}}\|^2$, $\sum \|O^{\mathrm{full}}\|^2$, and the query count, in float64, summed over image queries and channels (`sca/calib.py`). The sampler's trajectory is that of unmodified full attention; the emulated operators only observe. Statistics are written per prompt and merged by summation, so prompts can be added later and split-half fits need no extra pass. Text-query outputs are not in the objective, because SCA sends text queries through the global branch only.

Geometry at emulated scale. At 1024^2 the grid is 64×64. Emulated scale 2 uses a 32×32 window, 2×2 subsampling onto a 32×32 grid, and 4×4 query tiles; emulated scale 4 uses a 16×16 window, 4×4 subsampling onto a 16×16 grid, and 2×2 tiles. At 1280^2 the grid is 80×80, and emulated scale 2 uses a 40×40 window, 2×2 subsampling onto a 40×40 grid, and 5×5 tiles.

Noise bins. With 28 steps and $\mu = 1.15$, σ runs from 1.0 to 0.105, so the lowest of the 14 bins is empty and copies its neighbour; 13 bins are populated. Under this schedule 8 steps have $\sigma > 0.89$ (the ninth has $\sigma = 0.888$) and 13 have $\sigma > 0.80$.

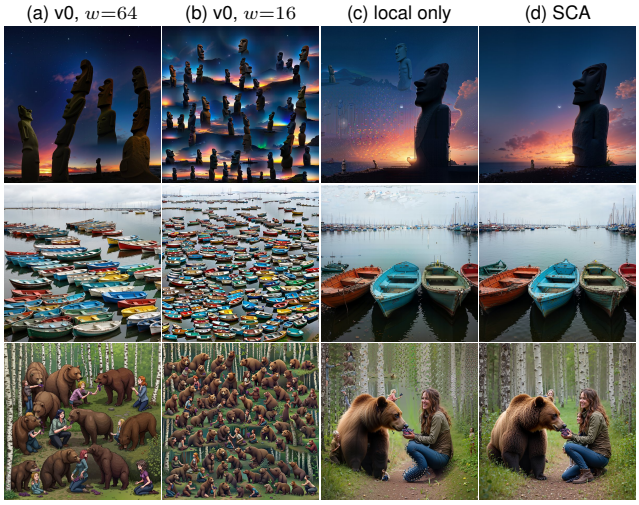

Figure 8: **Windows at high noise duplicate subjects.** 4096^2, three development prompts. (a) v0, the calibrated mix from the first step: each 64-token window settles the layout on its own. (b) A 16-token window gives smaller, more numerous copies. (c) With the opening phase but local-only attention afterwards ($\pi = 1$), ghost copies and grid speckle remain. (d) SCA.

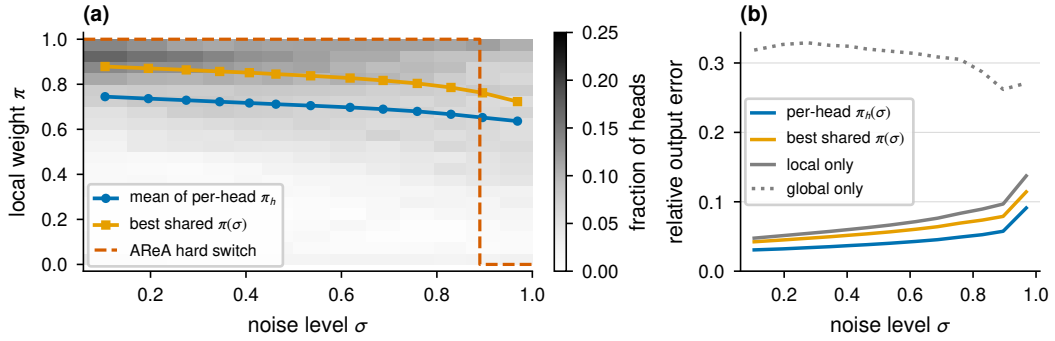

Figure 9: **In output energy, no noise level prefers the global branch.** (a) Distribution of $\pi_h(\sigma)$ over the 1368 heads per noise bin (grey), the mean over heads, and the best single shared weight, at emulated scale 2. The dashed step is AReA's schedule read as a weight. (b) Relative output error per bin. The local branch alone beats the global branch alone in every bin, and per-head mixing beats both. 32 prompts.

No global phase in output energy. The best single shared weight falls from 0.88 at $\sigma = 0.10$ to 0.72 at $\sigma = 0.97$, and the mean per-head weight from 0.74 to 0.64 (Figure 9). Among hard switches from global to local, the best threshold is $\sigma^\star = 1$, which equals local only. A switch at $\sigma^\star = 0.89$ between our two branches raises the error from 0.092 to 0.152 at scale 2 and from 0.261 to 0.376 at scale 4. In the noisiest bin, the global branch alone has relative error 0.27 at scale 2, the local branch alone 0.14, and the per-head mix 0.09. The generation results of Section 4.2 show the opposite preference for the opening steps, which is the gap between squared output error and layout.

Attention mass. For 256 random image queries per block and step, the pass also records the softmax mass on text keys and on image keys within Chebyshev distance 2, 4, 8, and 16. At 1024^2, averaged over heads and queries, text receives 15 to 19% of the mass; image keys within distance 16 receive 52% at $\sigma = 0.10$ and 40% at $\sigma = 0.97$, and keys outside that 33×33 neighbourhood still receive 29% and 41%. A purely local receptive field cannot account for any step even at native resolution.

Calibration budget. We merge the stored sufficient statistics from the first 4, 8, or 16 calibration prompts and rerun the 46 evaluation prompts with each resulting table. Table 17 reports generation-level differences from the 32-prompt table. The test reuses calibration observations but uses separate prompts for generation evaluation; it is not a held-out output-error benchmark.

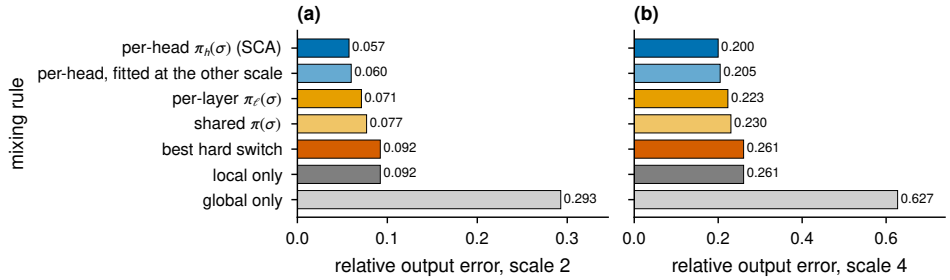

Figure 10: **Per-head weights explain full attention best, and transfer across scales.** Relative output error for each mixing rule at emulated scales 2 (a) and 4 (b), 32 prompts. "Fitted at the other scale" applies the scale-4 table at scale 2 and vice versa. The best hard switch puts its threshold at $\sigma^\star = 1$ and so equals local only. In-sample and held-out errors agree within 0.0003.

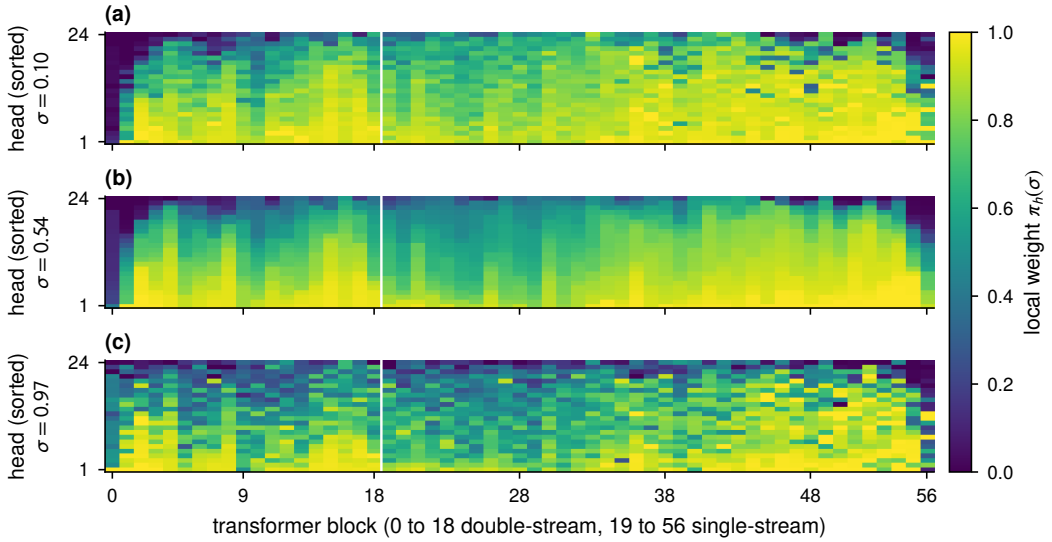

Figure 11: **Fitted weights at emulated scale 2.** $\pi_h(\sigma)$ for all 57 blocks and 24 heads of FLUX.1-dev at three noise levels, 32 prompts. Within each block, heads are sorted by their weight at $\sigma = 0.54$, in the same order in all panels. The white line separates double-stream from single-stream blocks. Figure 4 shows the scale-4 table used at inference.

C Implementation details and cost

FlexAttention window. Image tokens are reordered tile-major, so that a sparse block is a small spatial tile. Query blocks cover 8×16 tokens (128 queries) and key blocks 8×8 (64 keys), shifted so that unclamped windows start on a block edge; text keys come first. The mask function states window membership per query-key pair, so the result does not depend on the block layout, only the amount of skipped work does. On GH200, 8×16 query blocks with 8×8 key blocks were fastest at 2048^2, 4096^2, and 2048×6144. RoPE, subsampling, both branches, and the mix compile into one graph per layer; the mix is in fp32.

Numerical checks. The 18 CPU tests in `tests/test_sca.py` use float32 to check full-mode agreement with stock FLUX attention, both branches against brute-force Equations 2 and 3, full-grid and factor-1 identities, calibration when the local branch is exact, the gate, and FlexAttention against an explicit gather reference. A separate GPU check in `scripts/bench_attn.py` compares the gather and FlexAttention implementations in bf16: the recorded maximum absolute difference is 0.015625, and both have the same maximum error against the fp32 reference on the tested grids. This is an observed absolute-error check; no per-element ULP bound was measured.

Algorithm 1 SCA: calibration (once per model) and one attention layer at sampling time.

Calibrate(prompts, s_c) on an in-distribution $n \times n$ grid:
sample with full attention; at every block l, step σ, and head h:
  $O^{\mathrm{glob}}, O^{\mathrm{loc}} \leftarrow$ Eqs. 2, 3 with subsampling s_c, window n/s_c, tile $tn/(s_c n_0)$
  add $\|O^{\mathrm{loc}} - O^{\mathrm{glob}}\|^2$ and $\langle O^{\mathrm{full}} - O^{\mathrm{glob}}, O^{\mathrm{loc}} - O^{\mathrm{glob}} \rangle$ to $A_{lh\beta(\sigma)}$, $B_{lh\beta(\sigma)}$ ▷ float64
return $\pi_{lh\beta} \leftarrow \mathrm{clip}(B_{lh\beta}/A_{lh\beta}, 0, 1)$ ▷ Proposition 3
Layer(Q, K, V, σ) at target factors (s_H, s_W):
$O^{\mathrm{glob}} \leftarrow$ Eq. 2 for all queries ▷ block-centre keys, coarse positions
if $\sigma > \sigma_g$ **then return** O^{glob} ▷ opening phase
$O^{\mathrm{loc}} \leftarrow$ Eq. 3 for image queries ▷ one FlexAttention call
return $[\, O^{\mathrm{glob}}_{\mathcal{T}},\ \pi_h(\sigma)\, O^{\mathrm{loc}} + (1 - \pi_h(\sigma))\, O^{\mathrm{glob}}_{\mathcal{G}} \,]$ ▷ interpolated table; fp32 mix

Position ids in bf16. The FLUX.1 pipeline builds image position ids in bf16, which cannot represent every integer above 256. Our frozen runs retain these rounded ids for RoPE. Passing the true grid from the sampler fixes window membership and subsampling geometry, but does not restore exact RoPE coordinates: the coarse-query positions are also computed from the rounded ids. At a 384-token axis, index 383 rounds to 384 and its scale-6 coarse coordinate becomes 64; at a 512-token axis, index 511 rounds to 512 and its scale-8 coarse coordinate also becomes 64. A local 64-token window can likewise have a rounded endpoint difference of 64 instead of the ideal bound 63. Proposition 1 therefore applies exactly to the mathematical operator, not to these finite-precision positions at 2048×6144, 6144^2, or 8192^2. The 4096^2 grid uses indices 0–255, which bf16 represents exactly. FLUX.2 uses integer position ids. Rebuilding FLUX.1 positions in higher precision would require a separate controlled experiment; the reported runs are unchanged.

Non-divisible grids. When H/n_0 is not an integer, the subsampling factor per axis is the divisor of H nearest to H/n_0, and the tile side the divisor nearest to t; neither case arises at the sizes we evaluate.

Table 6: **Cost.** Top: analytic attention FLOPs per image (57 blocks, 24 heads, head dimension 128, 28 steps, $T = 512$); "gated" skips the local branch in the first 8 steps; ratio is full over gated SCA. Bottom: measured time of one attention layer in bf16 on GH200, minimum over interleaved runs; SCA is both branches and the mix.

Size ($H \times W$)	Grid	(s_H, s_W)	Full	SCA, no gate	SCA, gated	Ratio
2048×4096	128×256	(2, 4)	2.17×10^{16}	5.97×10^{15}	5.12×10^{15}	4.2
4096×4096	256×256	(4, 4)	8.56×10^{16}	1.19×10^{16}	1.02×10^{16}	8.4
2048×6144	128×384	(2, 6)	4.84×10^{16}	8.93×10^{15}	7.66×10^{15}	6.3

Token grid	Pixels	Full (SDPA, ms)	SCA (ms)	Speedup
128×128	2048^2	7.1	5.2	1.4
256×256	4096^2	87.3	18.7	4.7
384×128	6144×2048	49.6	14.0	3.5

End-to-end latency. The matched 4096^2 sweep on one GH200 gives medians of 186.5 s for full attention (186.0, 186.6, 186.5), 202.4 s for AReA (192.7, 202.4, 204.3), and 71.1 s for SCA (71.0, 71.1, 71.1). Each method generates one excluded warm-up and then three timed images in the same process, at batch 1 and 28 steps. The earlier benchmark, retained as a separate repeat, gave (`results/latency_4096_batch1.csv`): full attention 184.6 s (184.1, 184.7, 184.6), AReA 212.6 s (207.3, 225.8, 212.6), SCA + band guard 88.1 s (88.1, 98.7, 88.1; single-pass compiled band), and SCA final 71.0 s (71.0, 83.1, 70.9). On the shared campaign nodes, per-image times were 71 s for SCA final, 120 s for SCA + band guard, and 262 s for AReA; the campaign code used an unfused band, so dropping it saved more there (48 s) than in the clean measurement (17 s). At the non-square sizes we have only campaign timings on shared nodes (mean over 191 prompts): 2048×4096, SCA 38.1 s and AReA 112.4 s; 2048×6144, SCA 55.8 s and AReA 150.5 s. The two methods ran on the same pool of nodes. The continuing sweep adds clean 2048^2 medians of 22.9 s for full attention, 41.0 s for AReA, and 18.2 s for SCA, and 3072^2 medians of 73.9, 114.3, and 40.2 s,

respectively; SCA takes 161.6 s at 6144^2 and 287.6 s at 8192^2. Each is the median of three images after a generated warm-up in the same process (`results/latency_curve_summary.csv`). At 6144^2 and 8192^2, AReA takes 482.3 and 952.0 s, respectively: SCA reduces measured time by 66.5% and 69.8%. [TODO: E3: finish the full-attention 6144^2 timing group, Table 3.]

D HYPERPARAMETERS

Table 7: **All hyperparameters** of the main runs. Code commit `2d11740`; `scripts/sample.py -method sca -pi-table calib/k1_1024.npz -gate-sigma 0.89`.

Group	Setting	Value
Model	checkpoint, precision	FLUX.1-dev, bf16 weights and activations
	blocks, heads, head dim	19 double-stream + 38 single-stream, 24, 128
	native grid n_0, text tokens T	64 (1024^2 pixels, 16 pixels per token), 512
Sampler	steps, solver	28, flow-matching Euler
	guidance, shift	4.5 (distilled guidance), $\mu = 1.15$ at every size
	seed, decoding	seed 0 per prompt, shared across methods; tiled VAE decoding
Local branch	tile t, window w	8, 64 (clamped at borders), original positions
	kernel	FlexAttention; query blocks 8×16, key blocks 8×8, tile-major order
Global branch	subsampling	block centre, factor (s_H, s_W) = target grid / n_0
	positions	queries at $c(p)$, keys at block index, text at 0
Mixing	table	per (block, head, σ bin), $57 \times 24 \times 14$, scale-4 entry
	interpolation	linear in σ between the mean sampled noise levels of occupied bins, constant outside
	gate	$\sigma_g = 0.89$ (8 of 28 steps global only)
Calibration	prompts, resolution	32 at 1024^2 (16 more at 1280^2 for transfer)
	emulated scales	2 and 4, one pass; 14 bins of width $1/14$
	hardware	one GH200; complete calibration runtime was not retained
Band guard	(not in final method)	average-pool $p = 16$, bilinear upsampling

E EVALUATION PROTOCOL AND METRIC DEFINITIONS

Prompts and references. The Aesthetic-4K evaluation split (Zhang et al., 2025a) has 195 prompts with 4K reference images. Four (boats, Moai, woman and bear, underwater surfer) were used to choose the opening phase and window and are excluded from generation evaluation, leaving 191. Distribution metrics retain all 195 official reference rows. The 46-prompt ablation subset is fixed before any ablation was run. HRGen-1K (Rajabi et al., 2026a) has 1000 prompts and no references; we use source rows 0–199, with ids `hrgen_0000`–`hrgen_0199`. The 48 calibration prompts were written for this purpose and share no prompt with Aesthetic-4K (the largest word-set Jaccard overlap with any evaluation prompt is 0.26).

Preference and alignment metrics. ImageReward v1.0 (Xu et al., 2023) (bicubic resize to 224, centre crop). HPSv2 and HPSv2.1 (Wu et al., 2023): cosine of normalized image and text features with the official validation transform (longest side 224, padded). PickScore (Kirstain et al., 2023): $\exp(\text{logit scale})$ times the cosine. CLIP score (Radford et al., 2021): $100 \max(\cos, 0)$ with ViT-L/14. Aesthetic: the LAION improved aesthetic predictor (sac+logos+ava1, linear MSE) on the L2-normalized CLIP ViT-L/14 image embedding (Schuhmann, 2022).

No-reference quality. MUSIQ (Ke et al., 2021) (pyiqa defaults) on the full image at native resolution, and MUSIQ_c, its mean over four fixed random 1024^2 crops per image; crop positions depend only on the prompt id, so they are identical across methods. CLIP-IQA (Wang et al., 2022) and ARNIQA (Agnolucci et al., 2024) (pyiqa defaults) on the image downscaled with antialiased bicubic resampling to a long side of 1024 pixels, the scale both were trained on.

Table 8: **Aesthetic-4K with FLUX.1-dev**, 191 prompts (100 at 6144^2). HPS, CLIP-IQA, and ARNIQA are $\times 100$; higher is better except latency and FID. Bold: best per size. * / † on full-attention and SCA rows: significantly better / worse than AReA (95% paired bootstrap interval excludes zero). The added baselines are paired against SCA in Appendix H. Latency at 4096^2: seconds per image, batch 1, idle GH200, median of three. ‡Mean campaign time on shared nodes, not directly interchangeable with clean latency. $^\circ$UltraImage uses its two-stage defaults. **?**: pending. Appendix F gives available intervals. Patch values use corrected reference crops. Distributional bootstrap policies are specified in Appendix E.

Method	Lat. (s)↓	IR	Aesth.	CLIP	PickS.	HPSv2	HPSv2.1	FID↓	FID_p ↓	MUSIQ	MUSIQ_c	CLIP-IQA	ARNIQA
4096×4096, scale factors (4, 4)													
Full attention	186.5	$-2.18^\dagger$	$4.51^\dagger$	$15.38^\dagger$	$17.94^\dagger$	$20.75^\dagger$	$7.61^\dagger$	$312.7^\dagger$	252.4	$22.9^\dagger$	$33.2^\dagger$	$49.0^\dagger$	$54.5^\dagger$
DyPE	$196.1^\ddagger$	-0.11	5.73	24.84	20.49	25.52	20.52	206.2	172.7	36.3	50.4	59.5	60.1
SEGA	$190.5^\ddagger$	0.55	6.03	26.93	21.64	27.15	24.37	165.4	126.0	37.4	50.4	66.8	64.8
UltraImage$^\circ$	$308.5^\ddagger$	**1.28**	**6.50**	28.80	**23.32**	**29.97**	**30.80**	142.7	110.6	33.5	41.8	72.7	**72.3**
AReA (released code)	202.4	0.92	6.28	27.67	22.35	28.64	27.20	146.0	103.4	39.4	50.4	67.5	67.3
SCA + band guard	88.1	1.20^*	6.46^*	28.87^*	22.90^*	29.40^*	28.95^*	142.3	**98.8**	41.4^*	61.0^*	70.9^*	70.8^*
SCA (final)	**71.1**	1.23^*	6.45^*	**29.11***	23.09^*	29.77^*	29.71^*	**138.4***	99.8	**45.3***	**61.2***	**72.9***	71.2^*
2048×4096, scale factors (2, 4)													
AReA (released code)	$112.4^\ddagger$	**1.13**	**6.41**	**28.51**	**22.67**	28.38	**27.66**	144.8	**112.8**	51.8	61.6	**74.8**	**73.0**
SCA (final)	**38.1**‡	**1.13**	$6.21^\dagger$	28.21	22.64	**28.55***	27.46	**143.8**	$118.3^\dagger$	**55.9***	**65.9***	$73.5^\dagger$	72.8
2048×6144, scale factors (2, 6)													
AReA (released code)	$150.5^\ddagger$	0.39	**6.15**	26.49	21.51	26.41	23.35	163.6	**106.5**	42.9	53.1	67.3	69.3
SCA (final)	**55.8**‡	**0.66***	$6.00^\dagger$	**27.01**	**21.74***	**26.54***	**23.48**	**143.8***	110.1	**48.4***	**60.6***	**69.0***	**70.2***
6144×6144, scale factors (6, 6), 100 prompts [TODO: E2]													
AReA (released code)	?	?	?	?	?	?	?	?	?	?	?	?	?
SCA (final)	?	?	?	?	?	?	?	?	?	?	?	?	?

Distribution metrics and reference correction. FID (Heusel et al., 2017) and KID compare 191 generated images with all 195 official reference rows, using Inception-v3 features (torch-fidelity) after bilinear resizing to 299^2. The reference rows contain 168 distinct image contents; repeated rows retain their official weights. Patch metrics use four 1024^2 crops per row (764 generated and 780 reference crops). An audit found that the original feature-bank assembly keyed reference crops by resolved image path, so repeated images reused the last row's crop features. We corrected the lookup to retain each row's intended cached crops. Whole-image FID/KID are unchanged; all patch point estimates reported here use the corrected assembly. Original CSVs remain archived for provenance. Inception score uses full images at 299^2, with 10 splits.

Intervals. Per-image intervals are 95% paired percentile bootstraps over prompt ids: original summaries use 1000 draws; separately recomputed comparisons use 2000 draws with seed 0. For the main SCA–AReA FID/KID and patch-FID/KID comparison, 1000 draws resample generated prompt ids jointly across methods and resample the 168 distinct reference-image identities, keeping all rows and crops from each sampled identity together. This retains the 195-row weighted point estimate while accounting for repeated reference images. The resulting FID difference is -7.63 $[-12.79, -2.94]$; patch FID is -3.60 $[-7.39, 1.15]$ and patch KID ($\times 10^3$) is -2.27 $[-4.73, 0.12]$. The earlier claim of a significant patch-KID improvement does not survive this corrected evaluation. The whole-image and patch FID/KID comparisons at both non-square sizes, the added baselines' patch comparisons, and the 56-step patch-FID comparison also use reference-image clusters. Other whole-image FID/KID intervals are labeled historical reference-row bootstraps; they have not been recomputed with image-identity clusters. Historical patch intervals are not attached to corrected estimates and are omitted where recomputation is incomplete. FID's finite-sample bias can place point estimates outside percentile intervals. Intervals are unadjusted for multiple comparisons and measure prompt/reference sampling uncertainty, not variability over independent model training runs.

F Extended results with intervals

The following tables are populated from archived evaluation CSV files. Per-image means were checked against the corresponding per-image records; distribution-level estimates use archived cached features, corrected reference-row crop assembly, and explicitly labeled bootstrap policies.

Table 9: **All metrics at** 4096^2, 191 prompts, paired against AReA (released code). Δ is method minus AReA with its 95% paired bootstrap interval (1000 draws); **blue** / *orange*: significantly better / worse than AReA. Set-level metrics (FID, KID and their patch versions, Inception score) are computed on the whole set; the final SCA FID/KID and patch-FID/KID intervals use reference-image clusters; other FID/KID intervals retain the historical reference-row bootstrap unless a corrected image-cluster interval is available. Inception-score intervals resample only generated images. Corrected patch values without recomputed intervals have no significance marks. ‡Mean campaign wall-clock on shared nodes, not a clean latency measurement (Table 3 gives clean latency). Sources: `results/final4k_nb_*`, `final4k_*`, `final4k_full_*`.

Metric	AReA	SCA final	Δ [95% CI]	SCA + band	Δ [95% CI]	Full attention	Δ [95% CI]
ImageReward	0.917	1.229	+0.312 [+0.240, +0.384]	1.198	+0.281 [+0.208, +0.358]	−2.178	−3.095 [−3.216, −2.978]
Aesthetic	6.28	6.45	+0.17 [+0.12, +0.22]	6.46	+0.18 [+0.12, +0.24]	4.51	−1.77 [−1.87, −1.67]
CLIP score	27.67	29.11	+1.45 [+1.06, +1.82]	28.87	+1.20 [+0.79, +1.61]	15.38	−12.28 [−13.10, −11.44]
PickScore	22.35	23.09	+0.73 [+0.62, +0.86]	22.90	+0.54 [+0.40, +0.71]	17.94	−4.41 [−4.61, −4.18]
HPSv2 (×100)	28.64	29.77	+1.12 [+0.95, +1.31]	29.40	+0.75 [+0.53, +0.99]	20.75	−7.89 [−8.23, −7.54]
HPSv2.1 (×100)	27.20	29.71	+2.50 [+2.17, +2.88]	28.95	+1.75 [+1.31, +2.22]	7.61	−19.59 [−20.39, −18.71]
MUSIQ	39.4	45.3	+5.8 [+4.1, +7.5]	41.4	+2.0 [+0.0, +3.9]	22.9	−16.5 [−18.0, −15.0]
MUSIQ_c	50.4	61.2	+10.9 [+9.5, +12.4]	61.0	+10.6 [+9.2, +12.3]	33.2	−17.1 [−18.8, −15.4]
CLIP-IQA (×100)	67.5	72.9	+5.4 [+4.0, +6.8]	70.9	+3.3 [+1.8, +4.9]	49.0	−18.6 [−20.4, −16.7]
ARNIQA (×100)	67.3	71.2	+3.9 [+3.0, +4.9]	70.8	+3.5 [+2.4, +4.6]	54.5	−12.8 [−14.3, −11.2]
Inception score	6.42	6.52	+0.09 [−0.47, +0.49]	6.10	−0.33 [−0.88, +0.18]	3.83	−2.60 [−3.14, −1.83]
FID↓	146.0	138.4	−7.6 [−12.8, −2.9]	142.3	−3.7 [−9.6, +1.0]	312.7	+166.7 [+145.2, +170.9]
KID↓ ($\times 10^3$)	7.11	6.09	−1.02 [−3.12, +1.16]	5.48	−1.63 [−3.84, +0.51]	163.97	+156.86 [+139.17, +174.94]
FID_p ↓	103.4	99.8	−3.6 [−7.4, +1.2]	98.8	−4.7	252.4	+149.0
KID_p ↓ ($\times 10^3$)	11.44	9.17	−2.27 [−4.73, +0.12]	9.67	−1.77	171.16	+159.72
Seconds per image‡ ↓	261.7	71.5	−190.2 [−203.0, −176.8]	120.0	−141.7 [−155.8, −127.1]	196.5	−65.2 [−79.2, −50.5]

Table 10: **All metrics at the non-square sizes**, 191 prompts each, SCA final paired against AReA (released code). Conventions as in Table 9. Corrected patch intervals use reference-image clusters where available; absent intervals are omitted. Sources: `results/final_2048x4096_*`, `final_2048x6144_*`.

	2048 × 4096, factors (2, 4)			2048 × 6144, factors (2, 6)		
Metric	AReA	SCA	Δ [95% CI]	AReA	SCA	Δ [95% CI]
ImageReward	1.133	1.130	−0.003 [−0.079, +0.073]	0.389	0.664	+0.275 [+0.150, +0.400]
Aesthetic	6.41	6.21	−0.21 [−0.26, −0.15]	6.15	6.00	−0.15 [−0.25, −0.04]
CLIP score	28.51	28.21	−0.30 [−0.70, +0.11]	26.49	27.01	+0.52 [−0.01, +1.07]
PickScore	22.67	22.64	−0.02 [−0.13, +0.09]	21.51	21.74	+0.23 [+0.05, +0.43]
HPSv2 (×100)	28.38	28.55	+0.17 [+0.06, +0.28]	26.41	26.54	+0.13 [+0.00, +0.27]
HPSv2.1 (×100)	27.66	27.46	−0.20 [−0.49, +0.05]	23.35	23.48	+0.13 [−0.27, +0.54]
MUSIQ	51.8	55.9	+4.1 [+2.7, +5.7]	42.9	48.4	+5.5 [+3.6, +7.3]
MUSIQ_c	61.6	65.9	+4.3 [+3.3, +5.5]	53.1	60.6	+7.5 [+6.2, +9.0]
CLIP-IQA (×100)	74.8	73.5	−1.3 [−2.3, −0.4]	67.3	69.0	+1.6 [+0.2, +3.1]
ARNIQA (×100)	73.0	72.8	−0.2 [−0.9, +0.5]	69.3	70.2	+0.9 [+0.1, +1.7]
Inception score	6.21	6.65	+0.44 [−0.16, +0.95]	5.36	6.35	+0.99 [+0.13, +1.22]
FID↓	144.8	143.8	−1.1 [−6.4, +4.3]	163.6	143.8	−19.8 [−24.4, −11.8]
KID↓ ($\times 10^3$)	7.95	7.87	−0.09 [−2.90, +2.97]	18.08	8.46	−9.62 [−13.94, −5.60]
FID_p ↓	112.8	118.3	+5.5 [+1.9, +9.5]	106.5	110.1	+3.6 [−0.2, +11.3]
KID_p ↓ ($\times 10^3$)	14.53	16.48	+1.95 [+0.39, +3.70]	13.66	15.36	+1.70 [−2.22, +6.07]
Seconds per image‡ ↓	112.4	38.1	−74.3 [−76.6, −72.0]	150.5	55.8	−94.7 [−96.9, −92.4]

G ADDITIONAL PROTOCOLS AND COMPLETED COMPARISONS

DyPE and SEGA. We use the existing baseline harness frozen at commit `c05ebdc`, with FLUX.1-dev, 28 steps, guidance 4.5, maximum shift, tiled VAE decoding, and the same prompt-derived seeds as SCA. These settings isolate the attention/position treatment on the checkpoint used in our main comparison. They do not reproduce the methods' reported FLUX.1-Krea-dev results; the separate Krea comparison remains pending. Their low FLUX.1-dev scores should therefore not be read as a ranking under each method's preferred checkpoint.

UltraImage. We run a guided two-stage configuration from the FLUX branch of the released implementation through `scripts/run_ultraimage.py`. It uses a 1024^2 first stage and a 4096^2 second stage, 30 steps per stage, guidance 3.5 then 7, the `super_imax` guidance schedule, shifted-

Table 11: **Ablation intervals for Table 5**, 46 prompts at 4096^2, paired against SCA final (2000 bootstrap draws). The first column gives SCA final's absolute value; the last row gives patch FID per variant (set-level, no paired interval). Sources: `results/ablnb_*`.

Metric	SCA final	$\pi = 1$	shared $\pi(\sigma)$	per-layer	$\pi = 0$	mean pool	v0, no gate
ImageReward	1.124	−0.357 [−0.524, −0.207]	−0.090 [−0.173, −0.012]	−0.054 [−0.159, +0.030]	−0.395 [−0.507, −0.295]	−1.518 [−1.847, −1.193]	−1.328 [−1.595, −1.064]
Aesthetic	6.45	−0.21 [−0.35, −0.07]	−0.09 [−0.18, −0.02]	+0.01 [−0.06, +0.07]	−0.76 [−0.87, −0.64]	−0.61 [−0.77, −0.44]	−0.08 [−0.17, +0.01]
CLIP score	29.41	−1.93 [−2.68, −1.20]	−0.38 [−0.78, +0.08]	−0.13 [−0.54, +0.28]	−3.48 [−4.43, −2.59]	−3.59 [−4.76, −2.49]	−1.65 [−2.58, −0.84]
PickScore	22.96	−1.20 [−1.44, −0.99]	−0.38 [−0.51, −0.26]	−0.16 [−0.28, −0.05]	−1.44 [−1.66, −1.22]	−2.60 [−3.00, −2.17]	−2.18 [−2.51, −1.85]
HPSv2 (×100)	29.71	−1.78 [−2.07, −1.50]	−0.96 [−1.19, −0.73]	−0.50 [−0.63, −0.37]	−3.32 [−3.72, −2.94]	−4.04 [−4.66, −3.43]	−1.89 [−2.34, −1.45]
HPSv2.1 (×100)	29.98	−2.86 [−3.35, −2.36]	−1.83 [−2.24, −1.43]	−0.70 [−0.98, −0.41]	−5.66 [−6.35, −4.98]	−9.75 [−11.30, −8.17]	−4.07 [−5.11, −3.04]
MUSIQ	46.1	−3.5 [−6.5, −0.7]	−4.9 [−7.9, −1.9]	+0.7 [−1.6, +3.0]	−3.5 [−6.7, −0.2]	−1.8 [−5.3, +1.9]	−5.6 [−9.0, −2.1]
MUSIQ$_c$	62.5	−0.3 [−2.5, +1.7]	−10.4 [−12.8, −8.3]	−2.4 [−3.6, −1.1]	−11.7 [−13.6, −9.8]	−4.2 [−8.4, −0.5]	+2.5 [−0.4, +5.5]
CLIP-IQA (×100)	71.3	−4.6 [−6.3, −3.0]	−7.7 [−9.4, −6.0]	−0.9 [−2.0, +0.1]	−27.0 [−30.2, −23.8]	−10.1 [−13.0, −7.1]	+2.9 [+1.0, +4.9]
ARNIQA (×100)	70.5	−1.0 [−2.0, +0.0]	−6.5 [−8.5, −4.8]	−1.5 [−2.3, −0.9]	−45.1 [−47.6, −42.5]	−4.8 [−6.7, −2.6]	+4.3 [+2.7, +6.1]
FID$_p$ ↓	168.0	192.2	166.7	167.7	237.0	184.6	202.6

Table 12: **Intervals for the band-guard ablations** (Table 21 and further variants), 46 prompts at 4096^2, paired against SCA + band guard, whose absolute values form the first column. Corrected patch FID is shown without intervals. Sources: `results/abl_*`, `route_*`.

Metric	Reference	no band	no gate	shared $\pi(\sigma)$	per-layer	$\pi = 1$
ImageReward	1.165	−0.041 [−0.109, +0.021]	−0.357 [−0.511, −0.222]	−0.032 [−0.084, +0.015]	−0.024 [−0.055, +0.004]	−0.015 [−0.075, +0.040]
Aesthetic	6.43	+0.01 [−0.05, +0.08]	−0.23 [−0.35, −0.11]	−0.06 [−0.13, +0.00]	+0.03 [−0.01, +0.06]	+0.03 [−0.02, +0.10]
CLIP score	29.01	+0.40 [−0.04, +0.92]	−1.15 [−1.82, −0.47]	−0.09 [−0.53, +0.31]	−0.03 [−0.28, +0.23]	+0.16 [−0.19, +0.53]
PickScore	22.71	+0.25 [+0.09, +0.41]	−0.86 [−1.13, −0.59]	+0.02 [−0.10, +0.14]	+0.01 [−0.06, +0.08]	+0.12 [+0.03, +0.22]
HPSv2 (×100)	29.18	+0.54 [+0.31, +0.75]	−0.49 [−0.87, −0.14]	−0.20 [−0.42, +0.01]	−0.08 [−0.19, +0.04]	+0.16 [+0.04, +0.29]
HPSv2.1 (×100)	28.81	+1.17 [+0.71, +1.64]	−1.39 [−2.18, −0.63]	−0.33 [−0.68, +0.01]	−0.01 [−0.19, +0.19]	+0.55 [+0.30, +0.79]
MUSIQ	43.1	+2.9 [+0.7, +5.7]	−0.7 [−3.6, +2.4]	+0.5 [−2.5, +4.0]	−0.3 [−1.9, +1.8]	+1.0 [−0.3, +2.6]
MUSIQ$_c$	62.1	+0.4 [−0.9, +1.6]	−1.0 [−3.4, +1.7]	−5.7 [−7.2, −4.5]	−1.5 [−2.4, −0.7]	+0.9 [+0.3, +1.4]
CLIP-IQA (×100)	69.6	+1.7 [+0.3, +3.2]	−0.6 [−2.7, +1.5]	−6.5 [−8.4, −4.7]	−1.2 [−2.1, −0.5]	+0.5 [−0.9, +1.9]
ARNIQA (×100)	69.6	+0.9 [−0.6, +3.1]	+1.6 [−0.4, +4.5]	−4.6 [−7.1, −2.9]	−1.2 [−2.0, −0.6]	+0.6 [−0.1, +1.4]
FID$_p$ ↓	166.7	+1.3	+8.9	−4.6	−1.3	+1.6

window patchification, proportional attention, RIFLEx positions, and attention decay. Each prompt also needs a 2+2-step block-average preparation pass. The reported campaign time includes preparation when not cached and the two-stage call; main-stage and preparation times are retained separately in sidecars. Each call reseeds both the generator and the global RNG, including the VAE sampling path, from the prompt id and seed label. Thus seed labels are paired, but the two-stage method does not share an identical latent trajectory with the one-stage methods. The wrapper preserves an upstream scheduler behavior: assigning `scheduler.config.shift` does not update the scheduler's internal shift in the installed diffusers version, so both stages retain shift 3.0. This is a reproduction caveat, not a newly tuned setting. The paper reports this system-level comparison separately from the common-sampler attention comparisons.

FLUX.2-dev. The model has 56 attention blocks and 48 heads per block. We reuse the two branches and calibration rule with a native grid of 64^2 tokens, local window 64, and query tile 8. The four-axis positions preserve the time and text axes while coarsening the two spatial axes. Calibration uses 16 prompts at 1024^2, 14 noise bins, and emulated scales 2 and 4; sampling uses the scale-4 table. Both full attention and SCA use 28 steps, guidance 4.0, seed label 0, and a shift fixed to the native sequence length, $\mu = 2.1514$. The pipeline's default extrapolated shift at 4096^2 would be about 11.55; we therefore do not interpret this comparison as a default-pipeline benchmark. The gate is 0.959 so that the first 8 steps use only global attention. The 50 evaluation prompts are the first 50 Aesthetic-4K prompts remaining after development-prompt exclusion. Prompt embeddings are cached for both methods; reported times include denoising and tiled VAE decoding, and exclude text encoding and model loading. At emulated scale 4, fitted per-head relative output error is 0.1790, versus 0.1889 for per-layer weights and 0.1924 for local-only attention; split-half correlation is 0.950 and cross-half error 0.1794. These are calibration diagnostics, not generation ablations of the individual components on FLUX.2.

H Additional paired intervals

All differences below are method minus the named reference. Positive values favor the method for the ten per-image metrics. These comparisons resample matched prompt ids, with 2000 draws and seed 0; they do not treat multiple crops as independent images. The HRGen-1K sample has 200 prompts, the two extra seeds and sensitivity experiments each have 46, FLUX.2 has 50, and the main baseline comparison has 191. Tables use the separately recomputed per-image CSVs; these

Table 13: **Band-guard ablations, continued.** Conventions as in Table 12.

Metric	Reference	route 0.5	route 0.3	mean pool	$\pi = 0$	v0
ImageReward	1.165	-0.051 [-0.095, -0.009]	-0.026 [-0.082, $+0.024$]	-0.848 [-1.083, -0.615]	-0.436 [-0.537, -0.336]	-1.369 [-1.633, -1.113]
Aesthetic	6.43	-0.02 [-0.08, $+0.04$]	$+0.02$ [-0.03, $+0.07$]	-0.39 [-0.54, -0.25]	-0.75 [-0.86, -0.63]	-0.07 [-0.15, $+0.03$]
CLIP score	29.01	-0.00 [-0.31, $+0.28$]	$+0.13$ [-0.23, $+0.51$]	-2.96 [-4.18, -1.67]	-3.08 [-3.98, -2.16]	-1.25 [-2.02, -0.40]
PickScore	22.71	$+0.04$ [-0.03, $+0.12$]	$+0.09$ [$+0.01$, $+0.17$]	-1.80 [-2.10, -1.49]	-1.19 [-1.41, -0.96]	-1.94 [-2.23, -1.65]
HPSv2 ($\times 100$)	29.18	$+0.02$ [-0.11, $+0.15$]	$+0.11$ [-0.01, $+0.24$]	-2.35 [-2.81, -1.94]	-2.79 [-3.18, -2.45]	-1.35 [-1.74, -0.95]
HPSv2.1 ($\times 100$)	28.81	$+0.18$ [-0.04, $+0.40$]	$+0.34$ [$+0.12$, $+0.58$]	-5.81 [-6.96, -4.74]	-4.48 [-5.17, -3.80]	-2.90 [-3.82, -2.01]
MUSIQ	43.1	$+1.2$ [-0.2, $+3.0$]	$+0.8$ [-0.4, $+2.5$]	-1.8 [-5.5, $+1.7$]	-0.6 [-4.4, $+3.4$]	-2.7 [-6.1, $+0.7$]
MUSIQ$_c$	62.1	$+0.5$ [-0.0, $+1.1$]	$+0.8$ [$+0.2$, $+1.4$]	-0.8 [-5.0, $+3.6$]	-11.3 [-13.3, -9.6]	$+2.9$ [-0.3, $+6.2$]
CLIP-IQA ($\times 100$)	69.6	$+0.3$ [-1.3, $+1.5$]	$+0.6$ [-0.7, $+1.7$]	-5.4 [-9.3, -1.6]	-25.2 [-29.1, -21.4]	$+4.6$ [$+2.1$, $+7.3$]
ARNIQA ($\times 100$)	69.6	$+0.1$ [-0.6, $+0.8$]	$+0.6$ [-0.1, $+1.3$]	-1.1 [-3.6, $+1.9$]	-44.3 [-47.4, -40.7]	$+5.1$ [$+2.8$, $+8.1$]
FID$_p$ $\downarrow$	166.7	$+1.5$	$+1.1$	$+18.7$	$+70.3$	$+35.9$

Table 14: **Calibration table at** 2048×4096, 46 prompts, paired against SCA final with the scale-4 table. Sources: `results/ps2_*`.

Metric	SCA, scale-2 table	AReA (released code)
ImageReward	$+0.012$ [-0.025, $+0.050$]	$+0.027$ [-0.141, $+0.192$]
Aesthetic	-0.00 [-0.03, $+0.02$]	$+0.29$ [$+0.16$, $+0.43$]
CLIP score	-0.02 [-0.22, $+0.17$]	$+0.19$ [-0.53, $+0.91$]
PickScore	$+0.05$ [$+0.00$, $+0.09$]	$+0.14$ [-0.09, $+0.39$]
HPSv2 ($\times 100$)	$+0.09$ [$+0.04$, $+0.13$]	-0.19 [-0.44, $+0.06$]
HPSv2.1 ($\times 100$)	$+0.17$ [$+0.08$, $+0.26$]	$+0.39$ [-0.21, $+1.01$]
MUSIQ	$+0.0$ [-0.5, $+0.5$]	-3.1 [-7.5, $+1.2$]
MUSIQ$_c$	-0.0 [-0.4, $+0.2$]	-4.1 [-6.7, -1.3]
CLIP-IQA ($\times 100$)	-0.6 [-1.2, $+0.0$]	$+1.3$ [-0.8, $+3.4$]
ARNIQA ($\times 100$)	-0.2 [-0.6, $+0.1$]	$+0.4$ [-1.5, $+2.8$]

intervals differ slightly from the original 1000-draw evaluator summaries. No multiplicity correction is applied.

I Sampling-budget and guidance checks

On the same 46 evaluation prompts, we compare the default 28-step SCA run with 56 steps at guidance 4.5 and with 28 steps at guidance 7. The 56-step run keeps the noise threshold at 0.89; it therefore changes the discretization of both the opening and mixed phases. It improves ImageReward by 0.097 $[0.039, 0.159]$, HPSv2.1 by 1.01, PickScore by 0.16, and crop MUSIQ by 2.26, but increases corrected patch FID by 7.05 $[1.16, 13.03]$ with reference-image cluster resampling. Increasing guidance to 7 alone lowers HPSv2.1 and PickScore and does not resolve an ImageReward difference. Thus the default schedule does not maximize every metric, and extra sampling can change the preference-quality balance.

Against UltraImage on this subset, 56-step SCA has ImageReward difference -0.038 $[-0.159, 0.078]$, HPSv2.1 difference -0.56 $[-1.21, 0.09]$, and PickScore difference -0.19 $[-0.40, 0.01]$. These intervals span zero and do not establish equivalence. Its MUSIQ advantage is 9.94 $[5.69, 13.98]$ and crop MUSIQ advantage 21.83 $[18.67, 24.73]$. Campaign means are 71.6 s for default SCA, 140.1 s for 56-step SCA, and 322.0 s for UltraImage on these 46 prompts. This is a sampling-budget sensitivity check, not an equal-FLOP or equal-wall-clock experiment: UltraImage has two resolutions, different guidance, and per-prompt preparation.

Pending comparisons. [TODO: E2: 6144^2 intervals.] [TODO: E9: Krea per-metric intervals.]

J Exploratory 8K comparison

Table 20 reports all 12 completed prompt pairs at 8192^2 under the shared FLUX.1-dev protocol. SCA has higher ImageReward, CLIP, PickScore, both HPS variants, and ARNIQA, with paired intervals excluding zero. Aesthetic score and the three remaining quality metrics have intervals spanning zero. In particular, full-image MUSIQ, crop MUSIQ, and CLIP-IQA have point estimates favoring AReA. These mixed quality scores and the grid and stripe failures in Figure 19 limit the evidence for reliable 8K generation. With only 12 prompts, this comparison is exploratory; we do not report distributional metrics for this set.

Table 15: **Additional baselines minus SCA** at 4096^2, 191 prompts. UltraImage uses its own two-stage defaults; DyPE and SEGA share the main sampler settings.

Metric	DyPE	SEGA	UltraImage
ImageReward	-1.334 [-1.478, -1.199]	-0.677 [-0.803, -0.562]	$+0.047$ [-0.016, +0.111]
Aesthetic	-0.72 [-0.81, -0.63]	-0.42 [-0.49, -0.35]	$+0.06$ [+0.01, +0.10]
CLIP	-4.27 [-4.89, -3.66]	-2.19 [-2.65, -1.77]	-0.31 [-0.63, -0.01]
PickScore	-2.60 [-2.79, -2.41]	-1.45 [-1.60, -1.31]	$+0.24$ [+0.14, +0.33]
HPSv2 ($\times 100$)	-4.25 [-4.55, -3.97]	-2.61 [-2.85, -2.37]	$+0.21$ [+0.05, +0.36]
HPSv2.1 ($\times 100$)	-9.18 [-9.95, -8.50]	-5.34 [-5.88, -4.81]	$+1.10$ [+0.80, +1.40]
MUSIQ	-9.0 [-11.1, -6.8]	-7.9 [-9.8, -6.0]	-11.7 [-13.6, -9.8]
MUSIQ_c	-10.9 [-12.7, -9.0]	-10.8 [-12.5, -9.2]	-19.4 [-20.7, -18.2]
CLIP-IQA ($\times 100$)	-13.4 [-15.3, -11.6]	-6.2 [-7.8, -4.6]	-0.2 [-1.3, +0.8]
ARNIQA ($\times 100$)	-11.1 [-12.5, -9.7]	-6.5 [-7.5, -5.4]	$+1.1$ [+0.4, +1.8]

Table 16: **Dataset, seed, and model checks.** SCA minus AReA for HRGen-1K and seeds 1 and 2; SCA minus full attention for FLUX.2-dev. All at 4096^2.

Metric	HRGen-1K	Seed 1	Seed 2	FLUX.2-dev
ImageReward	$+0.642$ [+0.539, +0.755]	$+0.305$ [+0.172, +0.458]	$+0.355$ [+0.244, +0.470]	$+0.456$ [+0.320, +0.598]
Aesthetic	$+0.26$ [+0.20, +0.33]	$+0.11$ [-0.00, +0.23]	$+0.16$ [+0.06, +0.25]	$+0.30$ [+0.18, +0.42]
CLIP	$+2.50$ [+1.99, +3.03]	$+1.51$ [+0.94, +2.12]	$+1.68$ [+1.05, +2.33]	$+0.38$ [-0.38, +1.12]
PickScore	$+1.18$ [+1.06, +1.31]	$+0.71$ [+0.51, +0.92]	$+0.86$ [+0.62, +1.11]	$+0.56$ [+0.29, +0.86]
HPSv2 ($\times 100$)	$+1.87$ [+1.67, +2.07]	$+1.02$ [+0.71, +1.31]	$+1.35$ [+0.99, +1.71]	$+0.85$ [+0.50, +1.21]
HPSv2.1 ($\times 100$)	$+4.48$ [+3.96, +4.98]	$+2.19$ [+1.63, +2.74]	$+2.80$ [+2.08, +3.47]	$+1.84$ [+1.10, +2.55]
MUSIQ	$+6.7$ [+4.7, +8.6]	$+5.0$ [+1.0, +9.1]	$+3.5$ [-0.9, +8.0]	$+16.0$ [+11.9, +19.9]
MUSIQ_c	$+12.7$ [+11.2, +14.3]	$+11.6$ [+9.2, +13.9]	$+11.0$ [+8.2, +14.3]	$+21.4$ [+18.2, +24.8]
CLIP-IQA ($\times 100$)	$+8.1$ [+6.5, +9.8]	$+2.2$ [+0.0, +4.5]	$+4.0$ [+1.7, +6.3]	$+6.6$ [+4.7, +8.6]
ARNIQA ($\times 100$)	$+7.0$ [+5.9, +8.0]	$+3.2$ [+1.9, +4.4]	$+3.8$ [+2.3, +5.4]	$+7.1$ [+5.4, +8.8]

All 8K scores were computed on CPU. HPS uses float32 here, whereas the main GPU evaluation uses mixed precision, so absolute scores across resolutions should not be interpreted as a controlled scaling comparison. Both methods within this table use the same evaluator settings. For native-resolution MUSIQ, attention queries are processed in chunks of 256 while retaining every key and native image patch. This bounds intermediate memory without changing the attention context; pre-trained equivalence checks at three smaller image sizes reproduced the unchunked scores exactly.

K Calibration table at non-square sizes

At $(s_H, s_W) = (2, 4)$ the two axes have different factors, and SCA uses the table of the emulated scale nearest the larger factor, scale 4. On the 46 ablation prompts at 2048×4096, the scale-2 table changes the paired metrics negligibly (Table 14): HPSv2 $+0.09$, HPSv2.1 $+0.17$ (both $\times 100$), and PickScore $+0.05$ are significant but small, and ImageReward ($+0.01$ $[-0.03, 0.05]$) and the other metrics are within noise. These small changes do not resolve the mixed SCA–AReA results at this size, and we keep the default rule.

L The low-band duplication guard

Construction. Let LP_p average-pool a head's image output over $p \times p$ tokens and upsample it bilinearly, and $\text{HP}_p = \text{I} - \text{LP}_p$. The guard replaces the low band of the mix M by that of the global branch:

$$O \;=\; \text{LP}_p\big(O^{\text{glob}}\big) + \text{HP}_p\big(M\big) \;=\; O^{\text{glob}} + \pi_h(\sigma)\,\text{HP}_p\big(O^{\text{loc}} - O^{\text{glob}}\big). \tag{6}$$

The local branch then contributes only its difference from the global branch above the cutoff. The cutoff should lie below $1/(2s)$, where the global branch aliases, and above $1/w$, below which the window cannot resolve; with $w = 64$ and $s = 4$ this leaves periods of 8 to 64 tokens. We used

Table 17: **Sensitivity to calibration budget and gate.** Variant minus default SCA, 46 prompts. Calibration budgets use the first 4, 8, or 16 of the 32 calibration prompts. The gate reference is 0.89.

Metric	4 prompts	8 prompts	16 prompts	$\sigma_g = 0.85$	$\sigma_g = 0.93$
ImageReward	$+0.012$ [-0.007, +0.029]	$+0.010$ [-0.004, +0.024]	-0.007 [-0.020, +0.005]	-0.015 [-0.098, +0.064]	-0.004 [-0.097, +0.069]
Aesthetic	$+0.01$ [-0.00, +0.02]	$+0.01$ [-0.00, +0.02]	$+0.00$ [-0.01, +0.01]	-0.19 [-0.26, -0.12]	$+0.00$ [-0.05, +0.06]
CLIP	$+0.08$ [-0.03, +0.19]	$+0.05$ [-0.04, +0.15]	$+0.04$ [-0.04, +0.13]	-0.27 [-0.82, +0.26]	-0.64 [-1.03, -0.29]
PickScore	-0.01 [-0.03, +0.01]	$+0.00$ [-0.01, +0.02]	$+0.01$ [-0.01, +0.03]	-0.12 [-0.27, +0.02]	-0.22 [-0.38, -0.07]
HPSv2 ($\times 100$)	$+0.02$ [-0.02, +0.05]	$+0.02$ [-0.00, +0.05]	$+0.00$ [-0.02, +0.03]	-0.48 [-0.69, -0.28]	$+0.02$ [-0.16, +0.20]
HPSv2.1 ($\times 100$)	-0.02 [-0.07, +0.04]	$+0.01$ [-0.03, +0.05]	-0.01 [-0.05, +0.04]	-0.85 [-1.22, -0.48]	-0.15 [-0.56, +0.24]
MUSIQ	-0.1 [-0.4, +0.3]	-0.1 [-0.5, +0.2]	$+0.1$ [+0.0, +0.3]	-1.9 [-3.5, -0.2]	$+0.4$ [-1.0, +2.0]
MUSIQ$_c$	$+0.1$ [-0.1, +0.3]	-0.0 [-0.2, +0.1]	-0.1 [-0.2, +0.0]	-2.0 [-3.0, -1.1]	$+0.6$ [-0.4, +1.9]
CLIP-IQA ($\times 100$)	$+0.2$ [-0.1, +0.4]	$+0.2$ [-0.1, +0.4]	-0.1 [-0.4, +0.1]	-3.2 [-4.8, -1.7]	$+0.0$ [-1.2, +1.1]
ARNIQA ($\times 100$)	$+0.1$ [-0.1, +0.3]	$+0.0$ [-0.1, +0.2]	-0.0 [-0.1, +0.1]	-4.0 [-5.9, -2.3]	$+1.8$ [+0.7, +3.1]

Table 18: **Distributional baseline differences**, method minus SCA on 191 prompts; negative favors the method. Whole-image FID/KID retain the historical 1000-draw reference-row bootstrap intervals. Patch estimates use corrected reference crops and 1000-draw reference-image cluster intervals. The whole-image intervals have not been recomputed with image clusters.

Metric	DyPE	SEGA	UltraImage
FID↓	$+67.8$ [+56.9, +79.3]	$+27.0$ [+21.0, +38.4]	$+4.4$ [+1.2, +10.5]
KID ($\times 10^3$)↓	$+41.49$ [+32.04, +52.45]	$+4.98$ [+0.95, +10.11]	$+0.31$ [-1.69, +2.12]
Patch FID↓	$+72.9$ [+59.6, +81.6]	$+26.2$ [+17.9, +34.0]	$+10.7$ [+6.5, +14.7]
Patch KID ($\times 10^3$)↓	$+49.57$ [+40.36, +59.66]	$+13.76$ [+8.95, +19.22]	$+7.48$ [+5.43, +9.83]

$p = 16$; $p = 8$ and $p = 32$ were worse on four development prompts, and a noise-dependent cutoff behaved like $p = 16$. FreeScale (Qiu et al., 2024) also fuses two self-attention outputs by frequency band, in the opposite direction: it keeps the blurred low band of attention over overlapping crops and takes the high band from full self-attention at the target size. ResDiT also combines global low frequencies and local high frequencies, using patch-wise FFT masks (Ma et al., 2025). Our optional guard uses pooled global attention and preserves the fitted per-head weight on the local-minus-global high-band residual. Its global-low/local-high construction is therefore not itself novel.

History: what it fixes and what it costs. On the development prompts, the band alone (no opening phase) fixed the boats but left extra statues and faces on the Moai and woman-and-bear prompts; the opening phase alone got the main subject right, but small extra figures appeared late on some prompts; both together gave one subject on the boats, Moai, and woman-and-bear prompts. This was our first full configuration. On 46 prompts, however, removing the guard improves HPSv2 ($+0.54$), HPSv2.1 ($+1.17$), PickScore ($+0.25$), CLIP-IQA ($+1.7$), and MUSIQ ($+2.9$), all significant, leaves the other metrics within noise, and saves 17 s per image on an idle GPU. The guard also hid what the per-head mix contributes. With it (Table 21), local only matched the per-head mix within 2% on preference metrics, because the guard replaces the low band of every head's output with that of the global branch whatever the head's weight, so it injects global low frequencies even at $\pi = 1$; without the guard local only loses on nearly every metric (Section 4.2). Likewise, with the guard one shared weight and per-layer weights matched the per-head table on preference metrics and lost only on no-reference quality, and hard per-head routing (10 to 17% of heads global at threshold 0.5, about 6% at 0.3) stayed within 2% of the mix. The guard is an option for prompts where late duplication is visible, not part of the default method.

M Reproducing AReA

We run AReA's released code with its FLUX.1 defaults (block-centre pooling, first phase for 8 of 28 steps, halo width 8, bridge ratio 0.5), the FLUX.1-dev checkpoint, guidance 4.5, 28 steps, μ at its maximum, and the seeds shared with the other methods. AReA's paper reports ImageReward 1.32 at 4K under its own settings; our shared protocol gives 0.92, so cross-paper numbers are not comparable, while our direct SCA–AReA comparison holds these settings fixed. UltraImage and FLUX.2 use the separate protocols documented above. AReA's paper reports 151 s for AReA and

Table 19: **Budget and guidance sensitivity**, 46 prompts. The first two columns compare with default SCA; the last compares 56-step SCA with UltraImage. Per-image differences and 95% paired intervals, 2000 draws.

Metric	56 vs. 28 steps	Guidance 7 vs. 4.5	56 steps vs. UltraImage
ImageReward	$+0.097$ [+0.039, +0.159]	$+0.019$ [-0.070, +0.113]	-0.038 [-0.159, +0.078]
Aesthetic	$+0.03$ [-0.02, +0.09]	-0.10 [-0.18, -0.03]	-0.03 [-0.12, +0.05]
CLIP	-0.32 [-0.92, +0.19]	-0.43 [-0.94, +0.01]	$+0.32$ [-0.43, +1.01]
PickScore	$+0.16$ [+0.02, +0.31]	-0.27 [-0.41, -0.14]	-0.19 [-0.40, +0.01]
HPSv2 ($\times 100$)	$+0.34$ [+0.13, +0.57]	-0.30 [-0.51, -0.10]	-0.02 [-0.34, +0.30]
HPSv2.1 ($\times 100$)	$+1.01$ [+0.59, +1.48]	-0.74 [-1.08, -0.38]	-0.56 [-1.21, +0.09]
MUSIQ	-1.3 [-3.5, +1.0]	$+1.3$ [-1.2, +3.7]	$+9.9$ [+5.7, +14.0]
MUSIQ_c	$+2.3$ [+1.2, +3.4]	$+1.0$ [-0.2, +2.3]	$+21.8$ [+18.7, +24.7]
CLIP-IQA ($\times 100$)	$+1.5$ [+0.2, +2.7]	-1.5 [-2.9, -0.1]	$+1.1$ [-1.4, +3.4]
ARNIQA ($\times 100$)	$+1.3$ [+0.3, +2.4]	-1.4 [-2.2, -0.7]	-0.8 [-2.9, +1.0]

Table 20: FLUX.1-dev at 8192^2, 12 matched prompts, seed 0. All metrics are higher-is-better. Intervals are paired prompt-bootstrap 95% percentile intervals (2,000 draws, bootstrap seed 0).

Metric	AReA	SCA	SCA − AReA [95% CI]
ImageReward	-0.376	0.675	$+1.051$ [$+0.386$, $+1.754$]
Aesthetic	5.64	5.72	$+0.08$ [-0.26, $+0.44$]
CLIP score	22.38	25.52	$+3.14$ [$+0.63$, $+6.11$]
PickScore	20.35	21.27	$+0.92$ [$+0.24$, $+1.75$]
HPSv2 ($\times 100$)	24.60	26.13	$+1.53$ [$+0.56$, $+2.66$]
HPSv2.1 ($\times 100$)	16.59	20.51	$+3.92$ [$+1.31$, $+6.85$]
MUSIQ	28.6	25.7	-2.9 [-6.8, $+0.8$]
MUSIQ_c	44.0	40.1	-3.8 [-10.4, $+1.6$]
CLIP-IQA ($\times 100$)	57.1	52.0	-5.1 [-10.5, $+0.6$]
ARNIQA ($\times 100$)	37.3	54.6	$+17.3$ [$+11.9$, $+22.6$]

314 s for full attention at 4096^2 on an H100; the matched sweep measures 202.4 s and 186.5 s on one GH200 (medians of three images). We use commit `9e38ec2` of the released repository, which includes a cost cap on halo and bridge tokens and a jagged halo edge; neither appears in the paper text, and the paper does not say whether its Table 1 used them. Recomputed from the code with 512 text tokens, the first-phase pooling factors are 4, 4, 4, 4, 2, 2, 2, 2 at 4096×4096 and 2048×4096, and 8, 8, 8, 4, 4, 4, 2, 2 at 2048×6144. The second-phase windows shrink over 20 steps from 180, 151, and 192 tokens respectively to 64. At 2048×6144 no window is cheaper than full attention once the text tokens are counted, so the first window falls back to 192 tokens and the cap drops the bridge and halo on the first three second-phase steps. Other differences from the release defaults are the FLUX.1-dev checkpoint instead of FLUX.1-Krea-dev, and one seed per prompt shared across methods instead of one seed for all prompts. [TODO: E9: AReA with its default FLUX.1-Krea-dev checkpoint, against recalibrated SCA.]

N Qualitative galleries

Galleries are uncurated unless stated: prompts are drawn at random from the evaluation prompts, excluding the development prompts and those shown in the main text.

O Failure cases and limitations

Late duplication. With the final method, small extra figures can appear on some prompts in the last 20 steps, when the local branch dominates most heads. The band guard suppresses some of them but costs preference scores.

Table 21: **Ablations around the band-guard configuration at** 4096^2, 46 prompts. The reference row gives absolute values for SCA + band guard (band 16); every other row changes one thing, keeps the guard unless stated, and reports its paired difference to the reference. Marks and units as in Table 5; corrected patch FID is shown without intervals.

Variant	Mix err.	IR	Aesth.	HPSv2	HPSv2.1	PickS.	CLIP	CLIP-IQA	ARNIQA	MUSIQ	MUSIQ$_c$	FID$_p$ ↓
SCA + band guard (reference)	0.200	1.17	6.43	29.18	28.81	22.71	29.01	69.6	69.6	43.1	62.1	166.7
no band (**= SCA final**)	0.200	−0.04	+0.01	+0.54	+1.17	+0.25	+0.40	+1.7	+0.9	+2.9	+0.4	+1.3
no gate	0.200	−0.36	−0.23	−0.49	−1.39	−0.86	−1.15	−0.6	+1.6	−0.7	−1.0	+8.9
one shared $\pi(\sigma)$	0.230	−0.03	−0.06	−0.20	−0.33	+0.02	−0.09	−6.5	−4.6	+0.5	−5.7	−4.6
per-layer $\pi_\ell(\sigma)$	0.223	−0.02	+0.03	−0.08	−0.01	+0.01	−0.03	−1.2	−1.2	−0.3	−1.5	−1.3
local only ($\pi = 1$)	0.261	−0.01	+0.03	+0.16	+0.55	+0.12	+0.16	+0.5	+0.6	+1.0	+0.9	+1.6
hard routing, threshold 0.5	n/a	−0.05	−0.02	+0.02	+0.18	+0.04	0.00	+0.3	+0.1	+1.2	+0.5	+1.5
hard routing, threshold 0.3	n/a	−0.03	+0.02	+0.11	+0.34	+0.09	+0.13	+0.6	+0.6	+0.8	+0.8	+1.1
mean pooling	n/a	−0.85	−0.39	−2.35	−5.81	−1.80	−2.96	−5.4	−1.1	−1.8	−0.8	+18.7

Articulated subjects. The underwater-surfer development prompt keeps doubled limbs in every variant; AReA's image shows a large translucent ghost.

Small factors. At 2048×4096 the window covers half the image height, so the local branch alone already sees most of the image along that axis, and the per-head mix gains little over AReA's shrinking windows; aesthetic score and corrected patch FID are significantly worse.

Larger scales and another backbone. The gallery sampled from the completed 12-prompt 8192^2 run still shows grid and stripe artifacts in both methods (Figure 19); this exploratory set does not establish general 8K quality. In FLUX.2-dev, improved aggregate metrics coexist with distorted facial geometry and fine texture in the randomly selected portrait example (Figure 20).

Text-image offsets. The local branch keeps out-of-range text-image offsets (Remark 1).

Metrics. No-reference quality metrics reward sharp repeated subjects (v0 scores higher on CLIP-IQA and ARNIQA), and FID against 195 references at 299^2 resolution sees little of the 4K detail; we therefore report every metric and the qualitative galleries.

P Compute and reproducibility

Compute. Generation and calibration used single NVIDIA GH200 GPUs on a shared cluster; the verified timing logs identify the GH200 120GB model. Some metric evaluations ran on CPUs. Surviving calibration logs do not establish its complete runtime. Generation at 4096^2 takes about 71 s per image for SCA, 262 s for AReA, and 197 s for full attention on shared nodes (campaign means); the main comparisons, ablations, and non-square sizes comprise more than 2,000 generated images. Surviving worker logs establish a lower bound of 123.3 GH200 worker-hours through 2026-10-17 06:59 UTC, including model loading, calibration, evaluation, and failed attempts. This bound unions 581 closed task intervals by physical node and conservatively accounts for timestamp precision; it measures occupied worker time, not GPU kernel activity. Missing historical logs, unmatched starts, and ongoing work prevent a complete campaign total. Overlapping scheduler allocation time and image-sidecar durations are not added to this bound.

Reproducibility. The code (attention processor, calibration, sampling, evaluation wrappers, AReA wrapper, and tests), the calibration table `k1_1024.npz`, the per-image metric files, and the per-comparison summary and paired-delta CSV files behind every table will be released. Main-text metric values and paired intervals are derived from those CSV files. SCA spool tasks ran from task-declared frozen code copies. Sidecars retain sampling settings and available code fingerprints; the inherited Krea baseline records do not establish one source revision for every image.

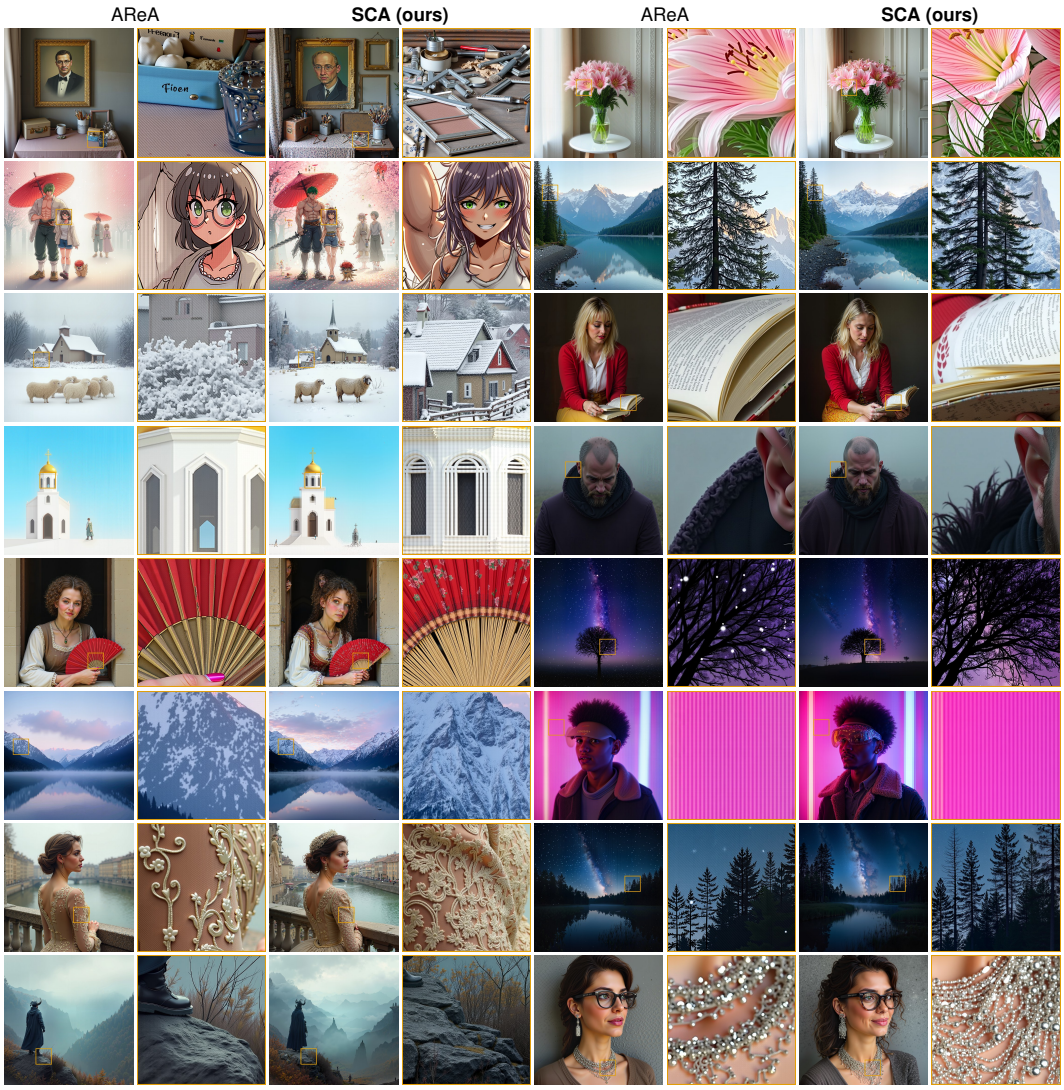

Figure 12: **Random prompts at 4096×4096, AReA vs SCA (ours).** 32 prompts sampled uniformly at random (seed 0) from the evaluation prompts, excluding the development prompts and those shown in the main text; not curated. Each full image is followed by a 1:1 crop of its outlined window, chosen automatically as the region with the most edge energy averaged over the methods. FLUX.1-dev, seed 0.

Q BROADER IMPACT

SCA shortens 4096^2 generation relative to full attention under our measured protocol, which helps applications such as print, design, and scientific visualization. Energy use remains unmeasured. The same reduction lowers the cost of producing high-resolution synthetic images that could be used to mislead. SCA adds no capability beyond the base model's and does not change its safety filters or watermarking; the risks and mitigations of the base model apply unchanged.

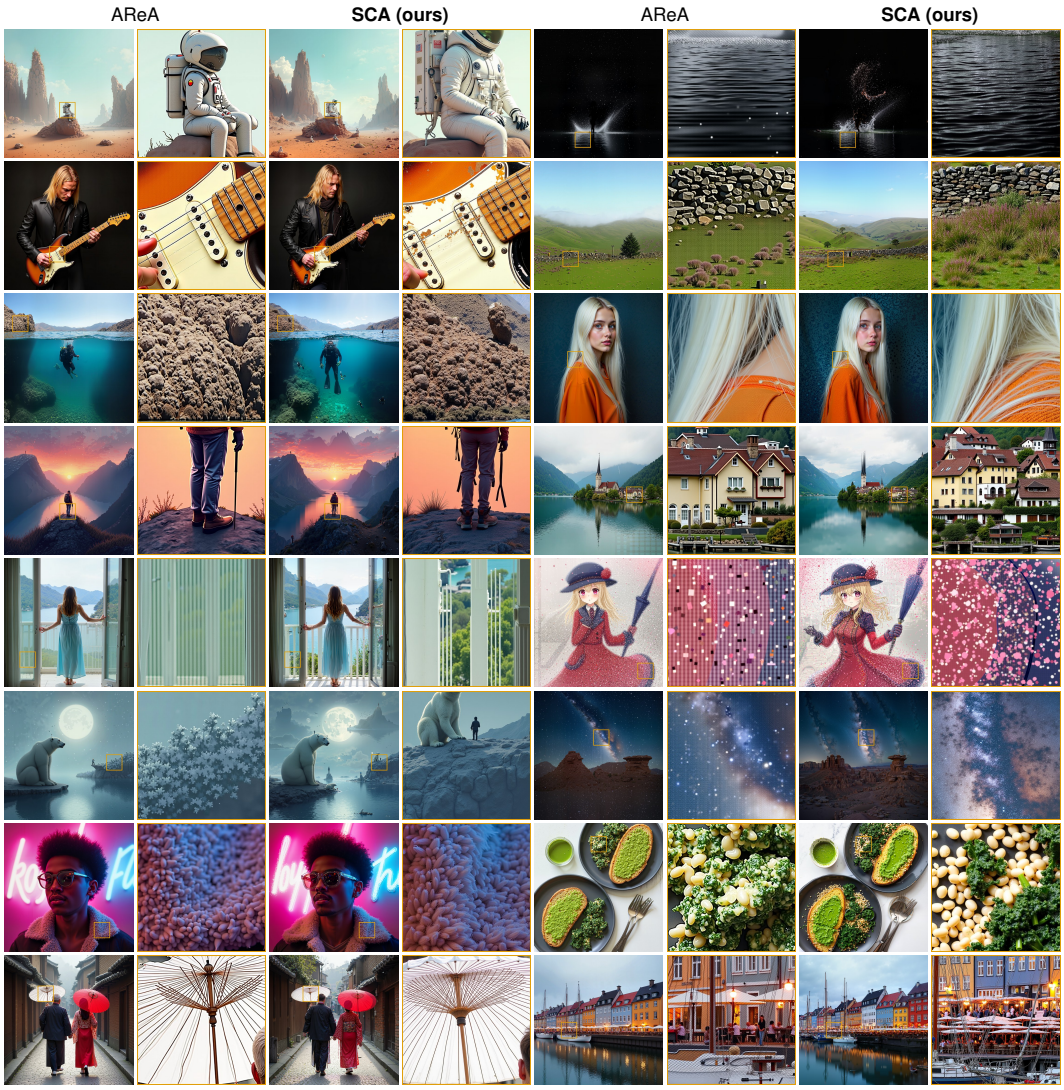

Figure 13: **Random prompts at 4096×4096, AReA vs SCA (ours) (continued).** Same protocol as Figure 12.

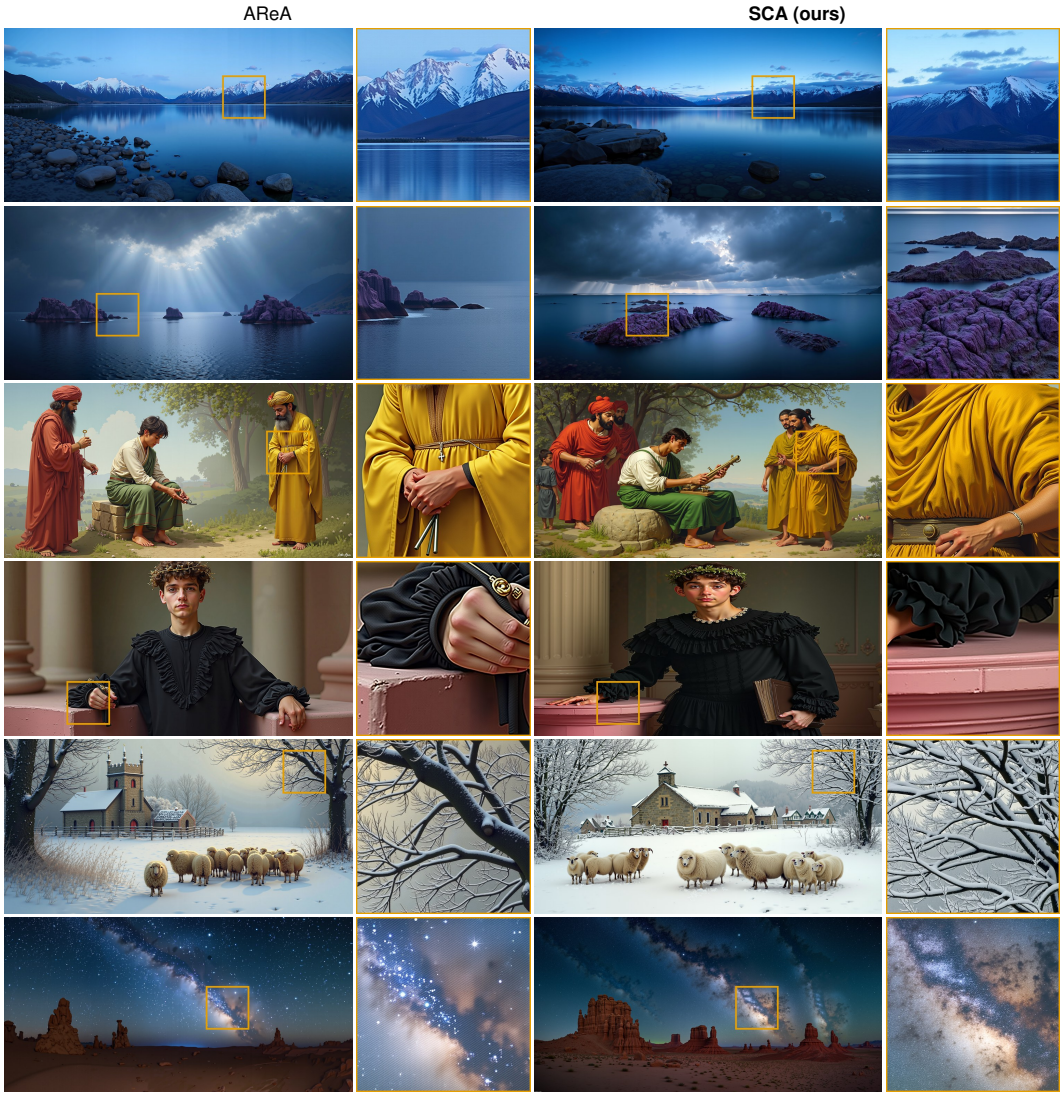

Figure 14: **Random prompts at 4096×2048 (width×height), AReA vs SCA (ours).** 6 prompts sampled at random from the evaluation prompts, disjoint from the other galleries and excluding the development prompts and those in the main text; not curated. Each image is followed by a 1:1 crop of its outlined 512×512 window, chosen automatically as the region with the most edge energy averaged over the methods. FLUX.1-dev, seed 0.

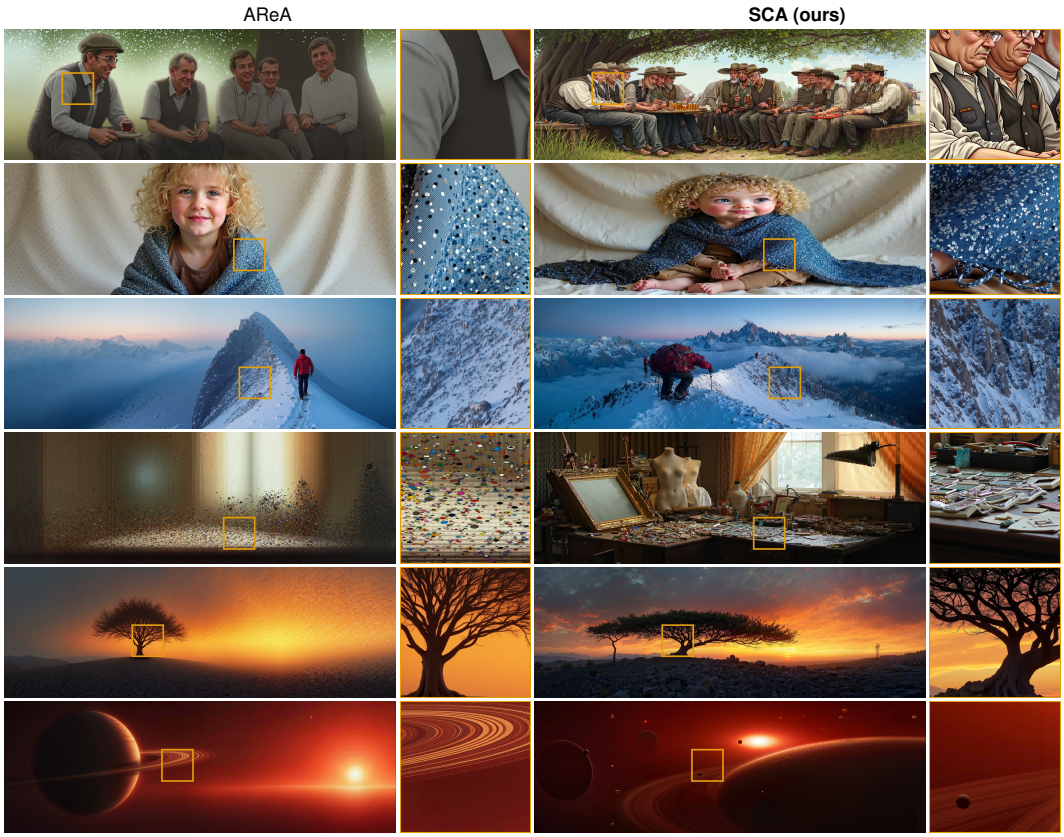

Figure 15: **Random prompts at 6144×2048 (width×height), AReA vs SCA (ours).** 6 prompts sampled at random from the evaluation prompts, disjoint from the other galleries and excluding the development prompts and those in the main text; not curated. Each image is followed by a 1:1 crop of its outlined 512×512 window, chosen automatically as the region with the most edge energy averaged over the methods. FLUX.1-dev, seed 0.

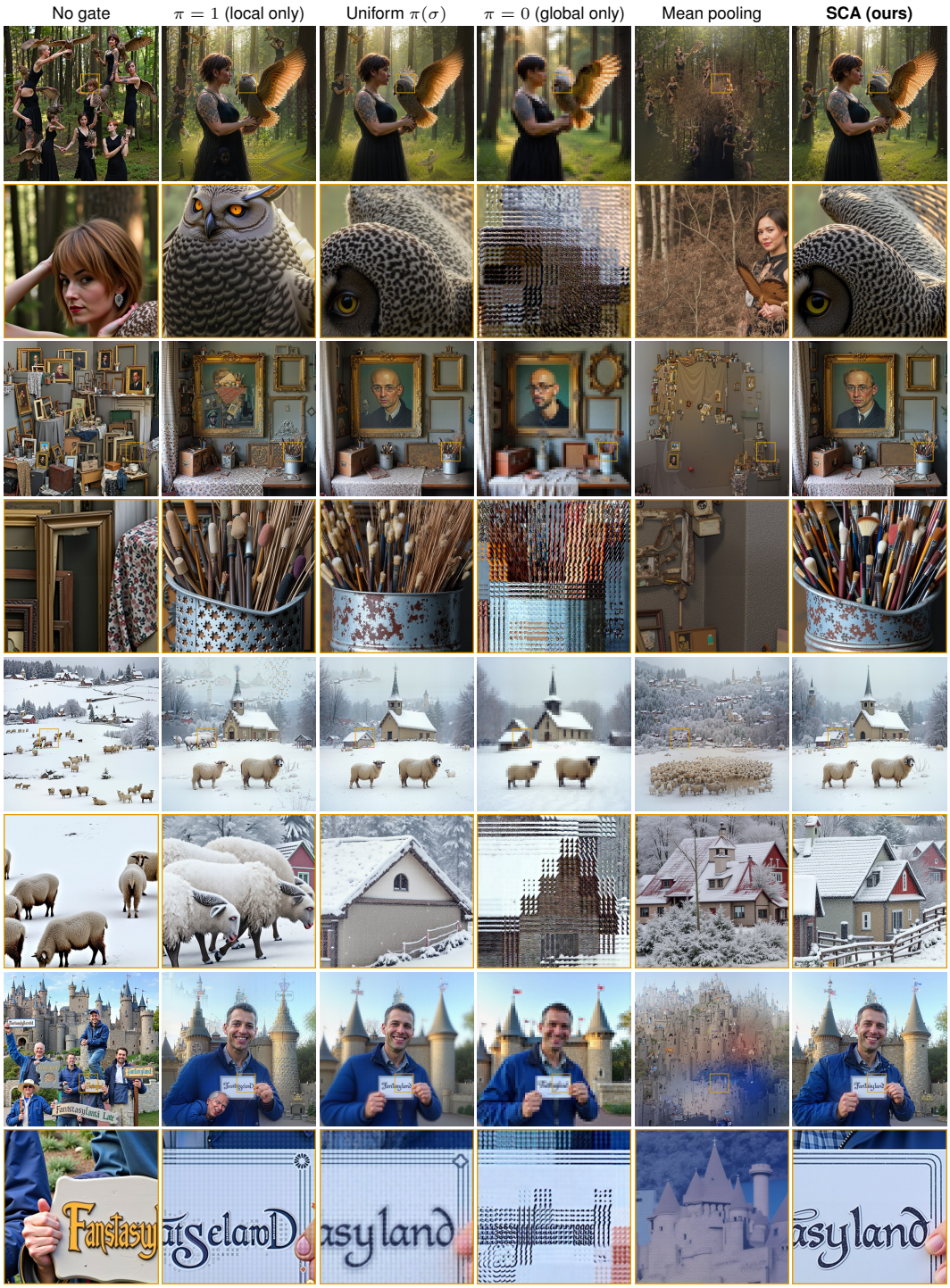

Figure 16: **Removing the gate duplicates subjects; dropping either branch or the per-head weights costs detail or coherence.** Four ablation prompts at 4096×4096, seed 0; each prompt shows the full images and, below them, 1:1 crops of the outlined 512×512 window (chosen automatically as the region with the most edge energy averaged over the six variants). No gate: the calibrated mix from the first step. $\pi = 1$ and $\pi = 0$: after the gate, the local or the global branch alone. Uniform $\pi(\sigma)$: one weight shared by all heads. Mean pooling: average-pooled instead of subsampled global keys. Without the gate the subject repeats across the canvas. $\pi = 0$ crops show the coarse grid of the pooled branch, and mean pooling loses the layout. $\pi = 1$ and uniform $\pi(\sigma)$ look close to SCA in these crops; their deficits show in the metrics.

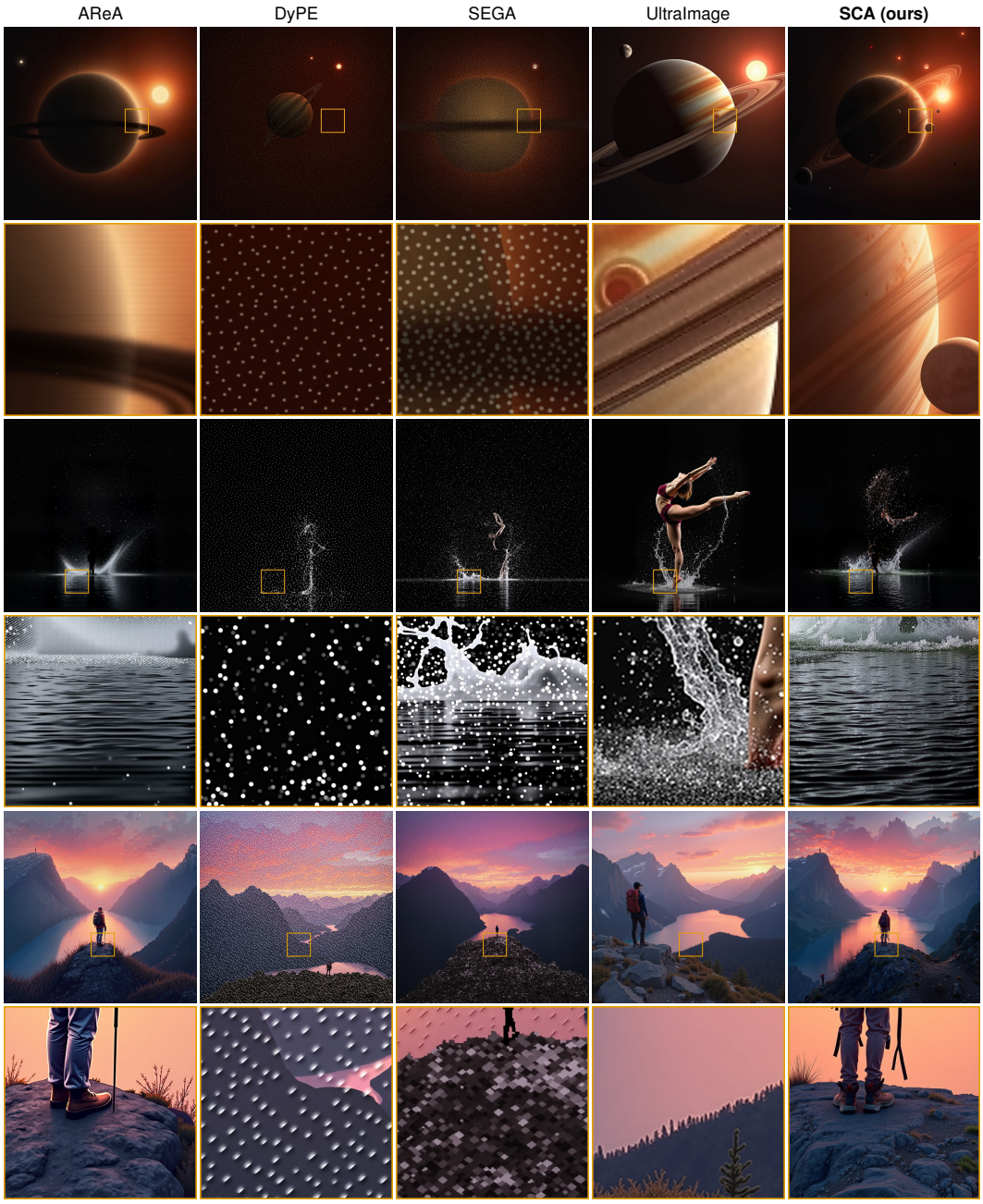

Figure 17: **Additional baselines at** 4096^2**.** Three randomly sampled paired prompts from the completed evaluation set, with no selection by visual quality or metric score. Each prompt occupies two rows: full images, then native-pixel crops of the outlined windows. FLUX.1-dev, seed 0. AReA, DyPE, SEGA and SCA use 28 steps and guidance 4.5; UltraImage uses its two-stage defaults (30+30 steps, guidance 3.5 then 7). The gallery illustrates outputs under these protocols, not equal-cost comparisons. Crop locations maximize edge energy averaged across the methods shown. Prompt selection uses seed 17; source hashes and crop coordinates are archived with the figure.

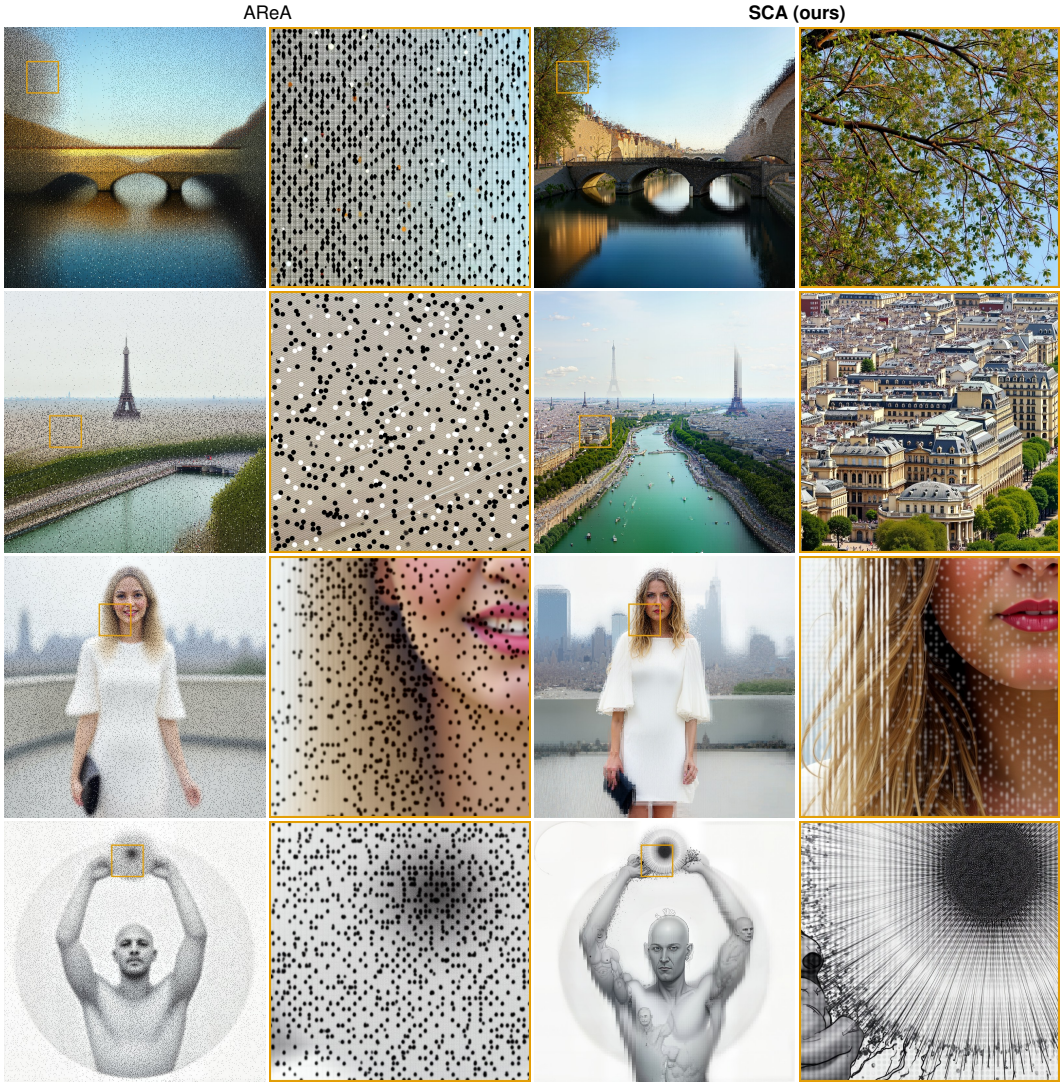

Figure 18: **Partial qualitative results at** 6144^2**.** 4 prompts sampled uniformly from the 14 non-development prompts with completed AReA and SCA outputs at the figure snapshot; this is not the final evaluation set. Each full image is followed by its native-pixel crop. FLUX.1-dev, seed 0, 28 steps, guidance 4.5. The subset was not selected by image quality; completion order may differ from the full benchmark. Crop locations maximize edge energy averaged across the methods shown. Prompt selection uses seed 17; source hashes and crop coordinates are archived with the figure.

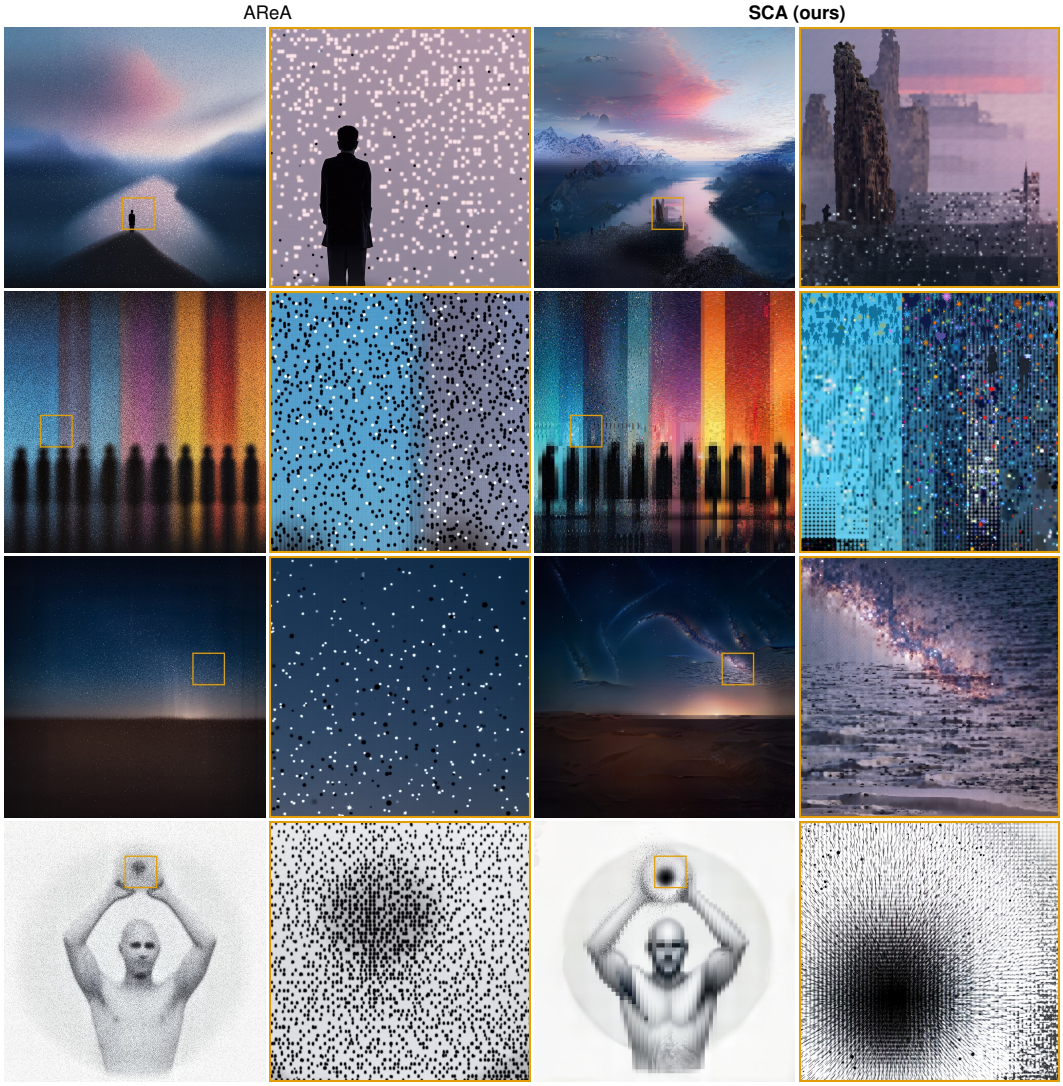

Figure 19: **Exploratory** 8192^2 **generation.** Four prompts sampled uniformly from all 12 paired evaluation prompts, without selection by visual quality or metric score. Each full image is followed by its native-pixel crop. FLUX.1-dev, seed 0, 28 steps, guidance 4.5. Grid and stripe artifacts remain at this scale; this small evaluation does not establish general 8K quality. Crop locations maximize edge energy averaged across the methods shown. Prompt selection uses seed 17; source hashes and crop coordinates are archived with the figure.

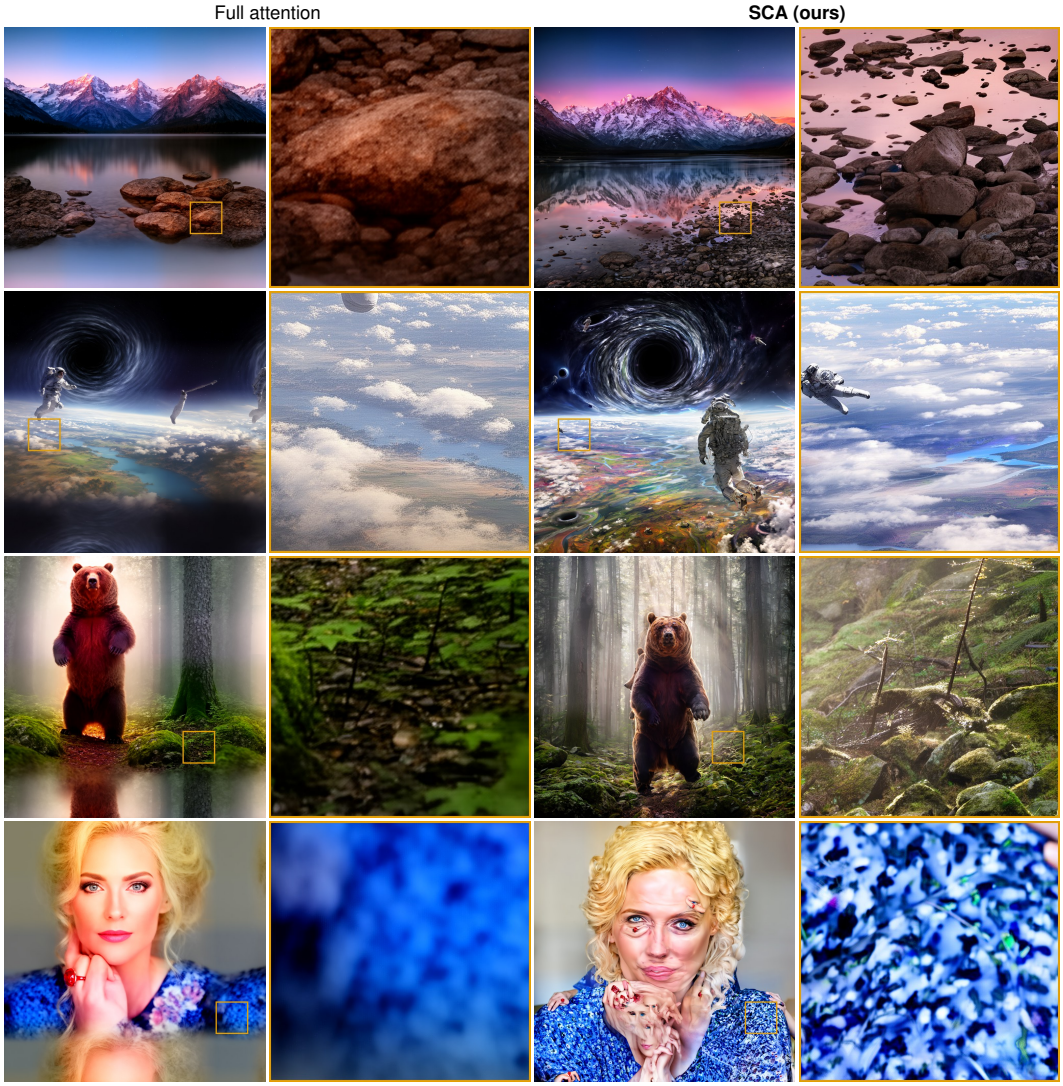

Figure 20: **FLUX.2-dev at** 4096^2**.** Four randomly sampled paired prompts from the completed 50-prompt evaluation, without quality-based selection. For each method, a full image is followed by its native-pixel crop. Seed 0, 28 steps, guidance 4.0, with the FLUX.2 calibration table for SCA. SCA can still distort face geometry and fine texture, as the last row shows. Crop locations maximize edge energy averaged across the methods shown. Prompt selection uses seed 17; source hashes and crop coordinates are archived with the figure.

Figure 21: **Failure modes.** Top: without the gate, local windows decide the layout at high noise and duplicate the subject (no gate, 64-token window; no gate, 16-token window, which gives smaller and more numerous copies); SCA keeps one subject. Two development prompts, 4096×4096. Middle and bottom: errors SCA still makes on evaluation prompts, with 1:1 crops of the outlined window and the same window from AReA. A thin diagonal sword is drawn as a staircase of short blocks; a patch beside the coat turns into a coarse grid; a dancer's rear arm ends in a foot; and the border of a dense illustration dissolves into speckle. AReA's images show none of these four errors, though its compositions differ, so its crops are not of the same objects. All seed 0.